# Chemistry — Class 12 (PCTB, English Medium)

## Solved Exercises | Punjab Curriculum and Textbook Board | Edition 2020

**NOTE: The Chapter 1, 2 and 3 exercises print no Q3 set (Q1, Q2, Q4 onward only). The book itself skips Ch.4 Q3(viii), Ch.8 Q3(viii) and Ch.16 Q11 (render-verified numbering skips, not omitted questions). All other previously flagged items are now resolved in place.**

## Chapter 1: Periodic Classification of Elements and Periodicity

### Exercise Q1 — Fill in the blanks

Mendeleev in his periodic table, arranged the elements according to their atomic masses .

Vertical columns in the modern periodic table are called groups and horizontal rows are called periods.

Members of group VIIA are called halogens and alkali metals is the family name of group IA members.

Metals form basic oxides and non-metals form acidic oxides.

Hydrogen can be placed above the groups IA, IVA and VIIA of the periodic table.

Shielding effect is actually the repulsion due to electrons in between the nucleus and the outermost shell.

Noble gases have the highest values of ionization energy due to their complete outermost shells.

When a second electron is added to a uni-negative ion, the incoming electron is repelled by the already present negative charge.

Due to having partly filled d-orbitals, transition metals usually show variable

valency.

Melting and boiling points of halogens increase down the group.

## Exercise Q2 — Indicate True or False

In Mendeleev's periodic table elements Be, Mg, Zn and Cd are placed in the same group.

True

The second and third periods contain eighteen elements each.

False

Alkaline earth metals are present in Group IIA.

True

Metals are present in the top right corner of the periodic table.

False

Metalloids are present in the lower half of Groups IVA, VA and VIA.

True

Hydrogen forms uni-negative ion like halogens.

True

Oxidation state of an element is related to the number of period it belongs.

False

Diamond is a good conductor of electricity.

False

Melting points of halogens decrease down the group.

False

Zinc oxide is an example of amphoteric oxide.

True

## Exercise Q4–Q10 — Short questions

Q4. What are the improvements made in Mendeleev's periodic table?

Mendeleev's table was improved when elements were arranged by atomic number instead of atomic mass.

This removed the wrong order of some pairs, like argon and potassium, and tellurium and iodine.

The modern table also gives separate places to isotopes and to the noble gases.

It explains periodicity clearly through electronic configuration and group numbers.

Q5. How does classification of elements in different blocks help in understanding their chemistry?

Block classification groups elements by the subshell being filled: s, p, d or f.

Elements of one block have similar outer electronic configurations, so their chemistry is similar.

It predicts valency, oxidation states and whether an element is a metal or non-metal.

Thus the whole chemistry of an element can be roughly predicted from its block and position.

Q6. How do you justify the position of hydrogen at the top of various groups?

Hydrogen resembles group IA metals: it has one outer electron and forms a positive ion.

It resembles halogens: it needs one electron for a stable shell and forms a negative ion.

It resembles group IVA elements: its valence shell is half-filled and it forms covalent bonds.

Because it matches no group fully, it is placed at the top, usually above group IA.

Q7. Why are the ionic radii of negative ions larger than the size of their parent atoms?

A negative ion is formed when an atom gains one or more electrons.

The number of protons stays the same, but the electrons now repel each other more strongly.

The electron cloud therefore expands and the effective nuclear pull on each electron falls.

So the anion is always larger than its parent neutral atom.

Q8. Why does ionization energy decrease down the group and increase along a period?

Down a group, new shells are added, so the outer electron is farther from the nucleus.

Inner electrons also shield the outer electron, so less energy is needed to remove it.

Along a period, electrons enter the same shell while the nuclear charge increases.

The stronger pull holds electrons tighter, so ionization energy increases along a period.

Q9. Why is the second value of electron affinity of an element usually shown with a positive sign?

The first electron affinity is exothermic because the neutral atom attracts the electron.

Adding a second electron means forcing it into an already negative ion.

The incoming electron is repelled, so energy must be supplied from outside.

Energy absorbed is shown with a positive sign, so the second electron affinity is positive.

Q10. Why does metallic character increase from top to bottom in a group of metals?

Metallic character depends on how easily an atom loses its outer electrons.

Down a group, atomic size increases and shielding by inner shells also increases.

The outer electrons are held less tightly by the nucleus.

They are lost more easily, so metallic character increases down the group.

Exercise Q11–Q14 (continued)

Q11. Explain the variation in melting points along the short periods.

Along a short period, melting and boiling points first increase from left to right up to the middle of the period and then decrease.

The increase is due to stronger metallic bonding as the number of delocalised electrons and nuclear charge increase.

After the middle, elements form discrete covalent molecules (for example P4, S8, Cl2) held by weak van der Waals forces, so melting points fall, and noble gases have the lowest values.

Q12. Why is the oxidation state of noble gases usually zero?

Noble gases have completely filled outermost shells (ns2np6, except helium 1s2), which is a very stable configuration.

They have very high ionization energies and positive electron affinities, so they neither lose nor gain electrons readily.

Their oxidation state is therefore usually zero, and they remain chemically inert under ordinary conditions.

Q13. Why is diamond a non-conductor and graphite fairly a good conductor?

In diamond, every carbon atom is sp3 hybridised and bonded to four others in a rigid three-dimensional network.

All valence electrons are used in covalent bonds, so there are no free or mobile electrons to carry current, and diamond is a non-conductor.

In graphite, each carbon is sp2 hybridised and bonded to three others in layers, with one delocalised pi electron per carbon free to move between layers, so graphite conducts electricity fairly well.

Q14. Give brief reasons for (a)-(h) as listed in the book.

(a) d- and f-block elements are called transition elements because their d or f subshells are partially filled in the atom or in a common oxidation state, giving them

transitional properties between s- and p-blocks.

(b) Lanthanide contraction is the steady decrease in atomic size across the lanthanides, caused by poor shielding by f electrons; it makes the 6th-period transition metals almost the same size as the 5th-period ones above them.

(c) Melting and boiling points rise up to the middle of the s- and p-blocks because metallic bonding strengthens with more delocalised electrons, then fall because the elements become molecular or gaseous with weak intermolecular forces.

(d) Oxidation states vary in a period because the number of valence electrons changes across the period, while in a group the valence electron count stays the same, so the common oxidation states remain almost constant.

(e) Hydration energy depends on charge density: Al3+ is small and highly charged, Mg2+ is intermediate, and Na+ is larger and singly charged, so hydration energy follows Al3+ > Mg2+ > Na+.

(f) Ionic character of halides decreases from left to right in a period because the metal becomes less electropositive and the bond becomes more covalent (Fajans' rules).

(g) Alkali metals are strongly electropositive and their hydrides are ionic: the metal loses its outer electron completely to hydrogen, giving M+ H-.

(h) Sodium and phosphorus are in the same period, but sodium is a metal and forms the basic oxide Na2O, while phosphorus is a non-metal and forms the acidic oxide P2O5, because metallic character decreases across a period.

Chapter 1 Q3 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 1 exercise pages as extracted show Q1, Q2 and Q4-Q14; no Q3 appears on those pages.

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# Chapter 2: s-Block Elements

## Exercise Q1 — Fill in the blanks

Alkali metals are more reactive than alkaline-earth metals.

Alkali metals decompose water vigorously producing alkalies and hydrogen.

When heated in a current of dry hydrogen, alkaline earth metals form white crystalline hydrides of the type MH2.

The beryllium hydroxide, like the hydroxide of aluminium, is amphoteric, while the hydroxides of the other members of group IIA are basic .

The elements of group IA are termed as alkali metals, because their hydroxides are alkaline.

Spodumene is an ore of lithium metal.

Alkali metal nitrates on heating give the corresponding nitrites and oxygen.

Na2CO3.H2O is the chemical formula of a mineral of sodium which is known as natron .

Metallic bicarbonates are decomposed on heating into their carbonates, along with water and carbon dioxide .

Metal nitrates other than the alkali metals on heating decompose into the corresponding metal oxides along with nitrogen peroxide and oxygen.

## Exercise Q2 — Indicate True or False

Group IA elements are called alkali metals because their chlorides are alkaline in nature.

False

Alkali metals are very good conductors of electricity.

True

The hydroxides of alkali metals and alkaline-earth metals are soluble in water.

False

Plaster of Paris is a hemihydrate.

True

Alkali metals have low melting and boiling points as compared to those of alkaline earth metals.

True

Lithium carbonate is decomposed to its oxide, but the carbonates of the other alkali metals are stable towards heat.

True

All alkali metal sulphates are insoluble in water.

False

Lithium combines with nitrogen to form lithium nitride but other alkali metals do not react with nitrogen.

True

Trona is a mineral of lithium.

False

Alkaline earth metals are stronger reducing agents than alkali metals.

False

**Exercise Q4, Q5, Q7, Q8, Q11 — Short questions**

Q4(a). Give the names, electronic configurations and occurrence of s-block elements.

Group IA (alkali metals): Li, Na, K, Rb, Cs, Fr, with outer configuration ns1.

Group IIA (alkaline-earth metals): Be, Mg, Ca, Sr, Ba, Ra, with outer configuration ns2.

They are very reactive, so they never occur free in nature.

They occur as chlorides, carbonates, sulphates and silicates in rocks, sea water and

minerals.

Q4(b). Discuss the peculiar behaviour of lithium with respect to the other members of alkali metals.

Lithium is very small in size with high charge density, so it behaves differently.

Its salts are less soluble, its carbonate and nitrate decompose on heating, unlike other alkali metals.

It forms a nitride with nitrogen directly, while other alkali metals do not.

It resembles magnesium diagonally, which is called diagonal relationship.

Q5. Discuss the trends in chemical properties of compounds like oxides, hydroxides, carbonates, nitrates and sulphates of IA and IIA group elements.

Oxides and hydroxides of both groups are basic, and their basic strength increases down the group.

Carbonates of alkali metals are stable to heat, but alkaline-earth carbonates decompose on heating.

Alkali metal nitrates give nitrites on heating, while other nitrates give oxides, NO2 and oxygen.

Sulphates of alkali metals are soluble, but sulphates of heavier alkaline-earth metals are insoluble.

Q7(b). Why is the aqueous solution of Na2CO3 alkaline in nature?

Sodium carbonate is a salt of a strong base, NaOH, and a weak acid, carbonic acid.

In water it hydrolyses to give sodium hydroxide and carbonic acid.

The free hydroxide ions make the solution alkaline.

That is why washing soda solution feels soapy and turns red litmus blue.

Q8(a). Describe with a diagram the manufacture of sodium by Down's cell.

In Down's cell, fused sodium chloride mixed with calcium chloride is electrolysed.

The mixture melts near 600 C, much lower than pure NaCl, which saves fuel.

At the iron cathode, sodium ions gain electrons and molten sodium collects.

At the graphite anode, chloride ions lose electrons and chlorine gas is released.

Q8(b). Point out the three advantages of this process.

First, the low melting mixture saves heat energy and cost.

Second, sodium and chlorine are obtained at the same time, both useful products.

Third, the cell runs continuously, so sodium is produced on a large scale.

Q11(a). Why are alkali and alkaline earth metals among the reactive elements of the periodic table?

They have only one or two electrons in their outermost shell.

These electrons are far from the nucleus and are shielded by inner shells.

Very little energy is needed to remove them, so the metals react readily.

They act as strong reducing agents in almost all their reactions.

Q11(c). How is gypsum converted into plaster of Paris?

Gypsum is calcium sulphate dihydrate, CaSO4.2H2O.

When it is heated to about 120 C, it loses three quarters of its water of crystallisation.

The product is calcium sulphate hemihydrate, CaSO4.1/2H2O.

This hemihydrate is plaster of Paris, which sets hard again when mixed with water.

Q11(e). Why is lime added to an acidic soil?

Acidic soil contains excess hydrogen ions which harm plant roots.

Lime, calcium oxide or calcium hydroxide, neutralises this acidity.

It supplies calcium, an important plant nutrient, to the soil.

It also improves soil structure, so air and water move better through it.

**Exercise Q6, Q7(a), Q9–Q11 (continued)**

Q6. Compare the chemical behaviour of lithium with magnesium.

Lithium resembles magnesium (diagonal relationship) because both are small with high charge density.

Both form normal oxides (not peroxides or superoxides) and nitrides by direct combination with nitrogen, while other alkali metals do not.

Their carbonates, hydroxides and nitrates decompose on heating (lithium carbonate to oxide, magnesium carbonate to oxide), and their chlorides are covalent and soluble in organic solvents.

Q7(a). Mention the properties of beryllium in which it does not resemble its own family.

Beryllium does not resemble its own family in several ways: it is very small with high charge density, so its compounds are largely covalent rather than ionic.

Its oxide and hydroxide are amphoteric, while other group IIA oxides and hydroxides are basic.

It does not react with water, does not form a stable carbonate, and forms complexes, resembling aluminium (diagonal relationship) rather than magnesium.

Q9(a). Compare the physical and chemical properties of alkali metals with those of alkaline earth metals.

Physical properties: alkali metals are softer, lighter, and have lower melting and boiling points than alkaline earth metals, because they have only one outer electron and weaker metallic bonding.

Both are good conductors of electricity and heat, but alkaline earth metals are harder and denser.

Chemical properties: both are very reactive and strong reducing agents, but alkali metals are more reactive; alkali metals form M+ ions and normal oxides, while alkaline earth metals form M2+ ions and are less vigorous with water.

Q9(b). What happens when: (i) lithium carbonate is heated, (ii) lithium hydroxide is heated to red hot, (iii) beryllium is treated with sodium hydroxide, (iv) lithium hydride is treated with water?

(i) Lithium carbonate on heating gives lithium oxide and carbon dioxide: Li2CO3

gives Li2O + CO2.

(ii) Lithium hydroxide heated to red heat gives lithium oxide and water: 2 LiOH gives Li2O + H2O.

(iii) Beryllium with sodium hydroxide gives sodium beryllate and hydrogen: Be + 2 NaOH gives Na2BeO2 + H2.

(iv) Lithium hydride with water gives lithium hydroxide and hydrogen: LiH + H2O gives LiOH + H2.

Q10. Give formulas of: (a) dolomite, (b) asbestos, (c) halite, (d) natron, (e) beryl, (f) sylvite, (g) phosphorite, (h) Chile saltpetre.

(a) Dolomite is CaMg(CO3)2 (CaCO3.MgCO3). (b) Asbestos is CaMg3(SiO3)4. (c) Halite is NaCl. (d) Natron is Na2CO3.H2O.

(e) Beryl is Be3Al2(SiO3)6. (f) Sylvite is KCl. (g) Phosphorite is Ca3(PO4)2. (h) Chile saltpetre is NaNO3.

Q11(b,d,f,g). Answer briefly: (b) lime water with CO2, (d) 2% gypsum in cement, (f) lime and sand for glass, (g) lime mortar.

(b) Lime water turns milky with carbon dioxide because insoluble calcium carbonate is formed: Ca(OH)2 + CO2 gives CaCO3 + H2O.

With excess carbon dioxide it becomes clear because the carbonate converts to soluble calcium hydrogen carbonate: CaCO3 + CO2 + H2O gives Ca(HCO3)2.

(d) 2% gypsum is added to cement to control its setting: it reacts with tricalcium aluminate and prevents flash setting, so the cement sets at a workable rate.

(f) Lime and sand are used to make glass: limestone (CaO), sand (SiO2) and soda ash are melted together to give soda-lime glass, a calcium sodium silicate.

(g) Lime mortar is prepared by mixing slaked lime, Ca(OH)2, with sand and water; it hardens by absorbing carbon dioxide from air to form calcium carbonate, which binds the sand.

Chapter 2 Q3 — flagged, not shown on the exercise pages.

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## Chapter 3: Group IIIA and Group IVA Elements

### Exercise Q1 — Fill in the blanks

Metaboric acid has the chemical formula HBO2.

Aluminium normally occurs as silicate minerals found in the rocks of the outer portion of the earth.

Aluminium gives sulphur dioxide gas with hot concentrated H2SO4.

Boron forms covalent bond in its compounds.

The chemical formula of white lead is 2PbCO3.Pb(OH)2 .

Boron is the only element with less than four electrons in the outermost shell that is not a metal.

In Group IIIA of the periodic table, boron is a semi-metal.

Borax that occurs as a natural deposit, is called tincal .

Cryolite is an important mineral of aluminium and its formula is Na3AlF6 .

A semiconductor is a substance that has different resistances to the passage of an electric current under different circumstances.

### Exercise Q2 — Indicate True or False

Boron always uses all the three of its valence electrons for bonding purposes.

True

Diaspore is an ore of carbon.

False

Emerald is an ore of aluminium that has the chemical formula AlO3.2SiO2.2H2O.

False

An aqueous solution of borax is feebly acidic in nature.

False

In the borax bead test of cupric oxide, the beads are coloured blue in the reducing flame.

False

Boric acid can be titrated with sodium hydroxide.

True

Carbon and silicon are the only non-metals in Group IVA.

True

PbO is commonly known as litharge.

True

Basic lead carbonate is a reddish brown pigment.

False

Aluminium oxide (Al2O3) is also called bauxite.

False

## Exercise Q4–Q11 — Short questions

Q4. What is the action of an aqueous solution of borax on litmus?

An aqueous solution of borax is alkaline in nature.

On hydrolysis it gives sodium hydroxide and boric acid.

The free alkali turns red litmus paper blue.

Blue litmus paper remains unchanged in borax solution.

Q5(a). Give the equation: borax is heated with CoO.

Borax first loses water and swells up, then melts to a clear glassy bead.

The bead is sodium metaborate and boric anhydride.

Cobalt oxide dissolves in the bead and forms cobalt metaborate.

CoO + B2O3 gives Co(BO2)2, which colours the bead deep blue.

Q6. Why is aluminium not found as a free element? Explain the chemistry of the borax bead test.

Aluminium is very reactive and has a strong attraction for oxygen.

In nature it is always found combined, mainly as oxides and silicates.

In the borax bead test, borax is heated on a platinum loop until it forms a glassy bead.

A coloured metal oxide dissolved in the bead gives a characteristic colour, which identifies the metal.

Q7. How does orthoboric acid react with: (a) sodium hydroxide (b) ethyl alcohol?

(a) With sodium hydroxide, orthoboric acid gives sodium borate and water.

H3BO3 + NaOH gives NaBO2 + 2H2O, and with excess alkali it gives borax.

(b) With ethyl alcohol in the presence of concentrated sulphuric acid it forms triethyl borate.

Triethyl borate burns with a green-edged flame, which is a test for boric acid.

Q9. Why are liquid silicones preferred over ordinary organic lubricants?

Liquid silicones keep their viscosity over a very wide range of temperature.

They do not freeze easily in cold and do not thin out much in heat.

They are chemically inert and are not attacked by water or oxidising agents.

They are also good electrical insulators, so they serve in machines and electrical parts.

Q10(a). CO2 is non-polar in nature. (b) CO2 is acidic in character.

(a) Carbon dioxide is linear, O=C=O, so the two C=O dipoles are equal and opposite.

They cancel each other, so the molecule has no net dipole moment and is non-polar.

(b) When CO2 dissolves in water it forms carbonic acid, H2CO3.

It turns blue litmus red and reacts with alkalies to form carbonates, showing acidic character.

Q11. Why is CO2 a gas at room temperature while SiO2 is a solid?

Carbon dioxide exists as small separate molecules held by weak van der Waals forces.

Very little energy is needed to separate them, so CO2 is a gas.

Silicon dioxide forms a giant covalent network in which every silicon is bonded to oxygen atoms.

Breaking this strong network needs great energy, so SiO2 is a hard solid with a high melting point.

**Exercise Q5(b), Q8, Q12–Q21 (continued)**

Q5(b). Give the equation: Al2O3 is heated with NaOH solution.

Aluminium oxide heated with sodium hydroxide solution gives sodium aluminate and water: Al2O3 + 2 NaOH gives 2 NaAlO2 + H2O.

The reaction shows the amphoteric character of aluminium oxide, which reacts with both acids and bases.

In solution the product is often written as sodium tetrahydroxoaluminate, Na[Al(OH)4].

Q8. How will you convert boric acid into borax and vice versa?

Boric acid is converted to borax by reacting it with sodium carbonate: 4 H3BO3 + Na2CO3 gives Na2B4O7 + CO2 + 6 H2O.

Borax is converted back to boric acid by acidifying a hot concentrated solution of borax with sulphuric acid, on cooling boric acid crystallises out: Na2B4O7 + H2SO4 + 5 H2O gives Na2SO4 + 4 H3BO3.

Q12. Give the names and the formulas of different acids of boron.

The acids of boron are: orthoboric acid, $H_3BO_3$; metaboric acid, $HBO_2$; pyroboric acid (tetraboric acid), $H_2B_4O_7$; and boric anhydride (boron sesquioxide), $B_2O_3$, which gives these acids with water.

Orthoboric acid is the common, stable acid, obtained from borax and from colemanite.

Metaboric acid is formed when orthoboric acid is heated to about 100 C, and pyroboric acid on stronger heating.

Q13. What is the importance of oxides of lead in paints?

Oxides of lead are important in paints as pigments and driers: white lead, $2PbCO_3.Pb(OH)_2$, is a white pigment with good covering power.

Red lead, $Pb_3O_4$, is used as an anticorrosive primer for iron and steel, because it protects the metal from rusting.

Litharge, PbO, is used in paints, in glass, and as a drier that speeds the drying of oil paints.

Q14. Give the names, electronic configurations and occurrence of Group IIIA elements.

Group IIIA elements are boron (B), aluminium (Al), gallium (Ga), indium (In) and thallium (Tl), with outer configuration $ns^2np^1$.

Boron occurs as borax and colemanite; aluminium is the most abundant metal in the earth's crust, occurring as bauxite, cryolite and feldspar.

Gallium, indium and thallium are rarer, occurring as traces in ores of zinc, aluminium and lead sulphides.

Q15. Discuss the peculiar behaviour of boron with respect to the other members of Group IIIA.

Boron behaves peculiarly compared with other group IIIA elements because it is very small and has no d orbitals, so its chemistry is covalent, not ionic.

It is a non-metal and a semiconductor, while aluminium and the heavier members

are metals.

Boron forms electron-deficient compounds (for example BF3, which accepts an electron pair) and does not form a stable B3+ ion, unlike the others which show the +3 oxidation state.

Q16. (a) What is borax? (b) Its commercial preparation. (c) Principal uses. (d) How does borax serve as a water softening agent?

Borax is sodium tetraborate decahydrate, Na2B4O7.10H2O, a white crystalline solid.

It is prepared commercially from colemanite (calcium borate) by boiling with sodium carbonate, which precipitates calcium carbonate and leaves sodium borate in solution, crystallised as borax; it also occurs naturally as tincal.

Its uses are in glass, glazes, detergents, antiseptics, and the borax bead test.

As a water softener, borax precipitates calcium and magnesium ions as insoluble borates, so soap is not wasted on hardness.

Q17. (a) What is boric acid? (b) Laboratory preparation. (c) Properties and uses.

Boric acid (orthoboric acid, H3BO3) is a weak, white crystalline acid.

In the laboratory it is prepared by acidifying a hot concentrated solution of borax with dilute sulphuric acid and cooling, when boric acid crystallises out.

Its properties are: it is a weak monobasic acid (it accepts OH- rather than donating H+ directly), it is soluble in hot water, and it gives a green flame with alcohol.

Its uses are as an antiseptic (boric lotion for eyes), in glass, glazes, and as a food preservative in small amounts.

Q18. (a) Three important ores of aluminium. (b) How aluminium reacts with oxygen, hydrogen, halogens, acids and alkalies.

(a) Three important ores of aluminium are bauxite (Al2O3.2H2O), cryolite (Na3AlF6) and feldspar (KAlSi3O8).

(b) Aluminium reacts: (i) with oxygen on heating to give aluminium oxide, 4 Al + 3

O2 gives 2 Al2O3; (ii) with hydrogen it does not react directly under ordinary conditions.

(iii) With halogens it gives aluminium halides, for example 2 Al + 3 Cl2 gives 2 AlCl3. (iv) With acids it gives a salt and hydrogen, for example 2 Al + 6 HCl gives 2 AlCl3 + 3 H2.

(v) With alkalies it gives an aluminate and hydrogen, for example 2 Al + 2 NaOH + 2 H2O gives 2 NaAlO2 + 3 H2.

Q19. Give the names, electronic configurations and occurrence of Group IVA elements.

Group IVA elements are carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb), with outer configuration ns2np2.

Carbon occurs free as diamond and graphite and combined in carbonates, coal and petroleum; silicon is the second most abundant element in the crust, in silica and silicates.

Germanium occurs in traces in zinc ores, tin as cassiterite (SnO2), and lead as galena (PbS).

Q20. Discuss the peculiar behaviour of carbon with respect to the other members of Group IVA.

Carbon behaves peculiarly compared with other group IVA elements because it is very small, has no d orbitals, and shows the strongest catenation (forming long chains and rings).

It forms multiple bonds (C=C, C triple C) readily, while the heavier members rarely do, and it is the only member that is a non-metal in all its common forms.

Carbon shows the +4 and -4 states and forms stable covalent compounds, while tin and lead show the +2 state prominently due to the inert pair effect.

Q21. (a) What are silicones? (b) Properties. (c) Uses. (d) What are silicates? (e) Uses of silicates.

(a) Silicones are organosilicon polymers with alternating silicon and oxygen atoms in the chain, with organic groups (usually methyl) attached to silicon.

(b) Their principal properties are: they are stable over a wide temperature range, water-repellent, chemically inert, and keep their viscosity in heat and cold.

(c) Their uses are as lubricants, water-repellent coatings, sealants, antifoams, and in cosmetics and medical implants.

(d) Silicates are salts of silicic acids, built from SiO4 tetrahedra joined in chains, sheets or frameworks, and are the main minerals of rocks and clays.

(e) Silicates are used in glass, cement, ceramics, zeolites (water softening and catalysis), and as fillers and building materials.

Chapter 3 Q3 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 3 exercise pages as extracted show Q1, Q2 and Q4-Q21; no Q3 appears on those pages.

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## Chapter 4: Group VA and Group VIA Elements

### Exercise Q1 — Fill in the blanks

The elements arsenic and antimony of group VA are called metalloids.

In Birkeland and Eyde's process, nitric acid is prepared from atmospheric oxygen and nitrogen.

The tendency to form long chains of atoms is called catenation .

All the elements of group VIA show the property of allotropy .

Selenium shows two allotropic forms which are called red and grey forms.

Specific gravity of H2SO4 at 18 C is 1.84 .

$H_2$ is produced by reacting $H_2SO_4$ with metals, like zinc .

The elements of group VIA exhibit maximum oxidation state of +6 .

The outermost shell of group VIA elements contain six electrons.

Oxygen shows paramagnetic behaviour due to the presence of unpaired electrons.

Conc. phosphoric acid acts as a dehydrating agent .

Nitrogen is a gas while other elements of the same group are solids .

Noble metals like gold and platinum are dissolved in aqua regia .

Sulphur is different from oxygen because it shows variable oxidation states.

$HNO_3$ is used in the manufacture of nitrogenous fertilizers.

## Exercise Q2 — Indicate True or False

The metallic character in groups VA and VIA elements increases down the group.

True

The elements of group VA exhibit maximum oxidation state of +5.

True

Ionization energy of phosphorus is greater than that of nitrogen.

False

The electronegativity of oxygen is greater than all other elements of groups VA and VIA.

True

$V_2O_5$ is used as a catalyst for the oxidation of $SO_2$ to $SO_3$.

True

The oxides of nitrogen are basic in nature.

False

Aqua regia is prepared by mixing 3 parts of conc. $HNO_3$ with one part of conc. HCl.

False

TNT is prepared by the reaction of nitric acid with toluene.

True

P2O3 when reacts with cold water gives phosphorous acid and with hot water it gives phosphoric acid.

True

Sulphur occurs in many organic compounds of animal and vegetable origins.

True

## Exercise Q3 — Multiple choice questions

Out of all the elements of group VA, the highest ionization energy is possessed by:

✓ A. N

B. P

C. Sb

D. Bi

Among group VA elements, the most electronegative element is:

A. Sb

✓ B. N

C. P

D. As

Oxidation of NO in air produces:

A. N2O

B. N2O3

✓ C. N2O4

D. N2O5

The brown gas formed when metal reduces HNO3 is:

A. N2O5

B. N2O3

✓ C. NO2

D. NO

Laughing gas is chemically:

A. NO

✓ B. N2O

C. NO2

D. N2O4

Out of all the elements of group VIA, the highest melting and boiling points are shown by:

✓ A. Te

B. Se

C. S

D. Pb

SO3 is not absorbed in water directly to form H2SO4 because:

A. the reaction does not go to completion

B. the reaction is quite slow

✓ C. the reaction is highly exothermic

D. SO3 is insoluble in water

Which catalyst is used in the contact process?

A. Fe2O3

✓ B. V2O5

C. SO3

D. Ag2O

Which of the following species has the maximum number of unpaired electrons?

✓ A. O2

B. O2+

C. O2-

D. O2 2-

## Exercise Q4 — Short questions

Q4(i). How does nitrogen differ from other elements of its group?

Nitrogen is a gas, while phosphorus, arsenic, antimony and bismuth are solids.

Its small size lets it form strong triple bonds, so N2 is very unreactive.

It cannot expand its octet because it has no d-orbitals, so its maximum covalency is four.

It forms no stable pentahalides, unlike phosphorus which forms PCl5.

Q4(ii). Why does aqua regia dissolve gold and platinum?

Aqua regia is a mixture of one part concentrated nitric acid and three parts concentrated hydrochloric acid.

Nitric acid oxidises gold to gold ions, while hydrochloric acid supplies chloride ions.

Gold ions combine with chloride to form the stable complex HAuCl4.

The complex formation drives the reaction forward, so even noble metals dissolve.

Q4(iii). Why do the elements of group VIA other than oxygen show more than two oxidation states?

Oxygen has no d-orbitals, so it shows mainly -2, -1 and +2 oxidation states.

Sulphur, selenium and tellurium have vacant d-orbitals in their valence shells.

Their paired s and p electrons can be promoted to d-orbitals for bonding.

They therefore show -2, +2, +4 and +6 oxidation states.

Q4(v). Write the equation for the reaction between conc. H2SO4 and copper, and explain what type of reaction it is.

Cu + 2H2SO4 gives CuSO4 + SO2 + 2H2O.

Copper is oxidised from 0 to +2, while sulphur in the acid is reduced from +6 to +4.

It is therefore an oxidation-reduction reaction.

Concentrated sulphuric acid here acts as an oxidising agent, not simply as an acid.

## Exercise Q4(iv), Q5–Q13 (continued)

Q4(iv). Write down a comparison of the properties of oxygen and sulphur.

Oxygen and sulphur differ in several ways: oxygen is a gas and sulphur is a solid at room temperature.

Oxygen shows mainly the -2 oxidation state (and -1 in peroxides), while sulphur shows -2, +2, +4 and +6, because it can use d orbitals.

Oxygen forms diatomic molecules (O2) with a double bond, while sulphur forms puckered S8 rings; oxygen is more electronegative and does not show catenation like sulphur.

Q5. (a) Birkeland and Eyde's process for nitric acid. (b) Which metals evolve hydrogen with nitric acid? (c) What is fuming nitric acid?

Birkeland and Eyde's process makes nitric acid from air: nitrogen and oxygen are heated to very high temperature (about 3000 C) in an electric arc, forming nitric oxide, N2 + O2 gives 2 NO.

The nitric oxide is cooled and oxidised by air to nitrogen dioxide, 2 NO + O2 gives 2 NO2, which is absorbed in water to give nitric acid and nitric oxide, 3 NO2 + H2O gives 2 HNO3 + NO.

(b) Most metals do not evolve hydrogen with nitric acid, because nitric acid is an oxidising acid; magnesium and manganese are exceptions and evolve hydrogen with very dilute nitric acid: Mg + 2 HNO3 (dilute) gives Mg(NO3)2 + H2.

(c) Fuming nitric acid is concentrated nitric acid containing dissolved nitrogen dioxide, which gives it a reddish-brown colour and fumes in air.

Q6. (a) Sulphuric acid as acid, oxidising agent and dehydrating agent (two reactions each). (b) Advantages of the contact process.

As an acid, sulphuric acid reacts with bases to give salts and water, and with carbonates to give a salt, water and carbon dioxide.

As an oxidising agent, it oxidises copper to copper sulphate with sulphur dioxide, and carbon to carbon dioxide with sulphur dioxide.

As a dehydrating agent, it removes water from sugar (charring it black) and from formic acid (giving carbon monoxide), and it absorbs water from gases.

Advantages of the contact process are: it gives a very pure, concentrated acid; sulphur dioxide is fully converted using a cheap catalyst (V2O5); and the process is continuous and economical on a large scale.

Q7. (a) Contact process from sulphur. (b) Why SO3 is dissolved in H2SO4 and not water. (c) Action of sulphuric acid on metals.

In the contact process, sulphur is burnt in air to sulphur dioxide: S + O2 gives SO2. Sulphur dioxide is oxidised to sulphur trioxide over vanadium pentoxide catalyst at about 450 C: 2 SO2 + O2 gives 2 SO3.

Sulphur trioxide is absorbed in concentrated sulphuric acid to give oleum, SO3 + H2SO4 gives H2S2O7, which is diluted with water to sulphuric acid of the required strength.

(b) SO3 is dissolved in H2SO4 and not in water because direct absorption in water is highly exothermic and forms a mist of acid that is difficult to condense; absorption in acid is smooth and controllable.

(c) With metals, dilute sulphuric acid gives a salt and hydrogen (for example Zn + H2SO4 gives ZnSO4 + H2), while concentrated hot sulphuric acid acts as an oxidising agent (for example Cu + 2 H2SO4 gives CuSO4 + SO2 + 2 H2O).

Q8. Describe the preparation of NO2 gas. Also give its reactions.

Nitrogen dioxide is prepared by heating a metal nitrate, for example 2 Pb(NO3)2 gives 2 PbO + 4 NO2 + O2, or by reacting copper with concentrated nitric acid, Cu + 4 HNO3 gives Cu(NO3)2 + 2 NO2 + 2 H2O.

Its reactions are: with water it gives nitric and nitrous acids, 2 NO2 + H2O gives HNO3 + HNO2; with alkalies it gives nitrite and nitrate; and it acts as a strong oxidising agent, for example oxidising sulphur dioxide to sulphur trioxide.

Q9. How can PCl3 and PCl5 be used for the preparation of other chemical

compounds?

Phosphorus trichloride is used to make phosphorus oxychloride, phosphorous acid, and to replace -OH groups by -Cl in organic compounds (for example alcohols to alkyl chlorides).

Phosphorus pentachloride is a stronger chlorinating agent: it converts acids to acid chlorides, alcohols to alkyl chlorides, and is used to test for hydroxyl groups (giving HCl fumes).

Both are important laboratory and industrial reagents for introducing chlorine into other compounds.

Q10. (i) Ring test for nitrate. (ii) NO2 as oxidising agent. (iii) HNO3 with arsenic and antimony. (iv) Preparation of PCl3. (v) P2O5 as dehydrating agent.

(i) Ring test: to a solution of the nitrate, add freshly prepared ferrous sulphate solution and then concentrated sulphuric acid carefully down the side of the tube; a brown ring of [Fe(NO)]SO4 forms at the junction of the two layers, confirming nitrate.

(ii) NO2 is a strong oxidising agent: it oxidises sulphur dioxide to sulphur trioxide, hydrogen sulphide to sulphur, and carbon monoxide to carbon dioxide, itself being reduced to nitric oxide.

(iii) With arsenic, concentrated nitric acid gives arsenic acid: As + 5 HNO3 gives H3AsO4 + 5 NO2 + H2O; with antimony it gives antimonic acid and nitrogen dioxide similarly.

(iv) PCl3 is prepared by passing dry chlorine through white phosphorus: P4 + 6 Cl2 gives 4 PCl3, or by heating phosphorus with sulphur chloride; it is purified by distillation.

(v) P2O5 is a powerful dehydrating agent: it removes water from nitric acid to give N2O5, from organic acids and amides, and it chars paper and sugar by removing the elements of water.

Q11. Complete and balance the chemical equations (i)-(x).

(i) 4 P + 10 NO gives 2 P2O5 + 5 N2. (ii) 2 NO + Cl2 gives 2 NOCl. (iii) H2S + 2 NO gives S + N2O + H2O.

(iv) 2 Pb(NO3)2 gives 2 PbO + 4 NO2 + O2. (v) 2 NO2 + H2O gives HNO3 + HNO2.

(vi) 2 NO2 + H2SO4 gives NOHSO4 + HNO3. (vii) 2 HNO2 + 2 HI gives 2 NO + I2 + 2 H2O.

(viii) HNO2 + NH3 gives N2 + 2 H2O. (ix) 2 HNO2 + CO(NH2)2 gives CO2 + 2 N2 + 3 H2O. (x) KNO3 + H2SO4 gives KHSO4 + HNO3.

Q12. Methods of preparation of phosphorus pentoxide and its reactions.

Phosphorus pentoxide is prepared by burning phosphorus in excess dry oxygen or air: P4 + 5 O2 gives 2 P2O5.

It is a white, deliquescent solid that reacts violently with water to give phosphoric acid: P2O5 + 3 H2O gives 2 H3PO4.

It is a powerful dehydrating agent: it converts nitric acid to dinitrogen pentoxide, amides to nitriles, and removes water from many organic compounds.

Q13. Discuss the trends in physical properties of group VIA elements.

In group VIA, atomic size and density increase down the group, while melting and boiling points generally increase (oxygen and sulphur are gases/solid with lower values, selenium and tellurium are solids with higher values).

Ionization energy and electronegativity decrease down the group, because the outer electrons are farther from the nucleus and more shielded.

Metallic character increases down the group: oxygen and sulphur are non-metals, selenium and tellurium are metalloids, and polonium is a metal; catenation and allotropy decrease down the group.

Chapter 4 Q3(viii) — flagged, not shown on the extracted exercise pages.

BOOK NUMBERING NOTE (render-verified): book p.41 jumps from Q3(vii) to Q3(ix); Q3(viii) is not printed. This is a book numbering skip, not an omitted question.

# Chapter 5: The Halogens and the Noble Gases

## Exercise Q1 — Fill in the blanks

The halogen with the highest electronegativity is fluorine .

HI is a strong reducing agent.

The only halogen acid which is not a strong acid is hydrofluoric acid .

Tincture of iodine is a common antiseptic.

The halogen with the lowest melting and boiling points is fluorine .

Bleaching powder is prepared from chlorine and slaked lime .

Bromine is the halogen which causes burns on the skin which heal slowly.

Chlorine is used for making unshrinkable wool.

A mixture of helium and oxygen is used for breathing by the sea divers.

Noble gas used to fill fluorescent tubes is neon .

## Exercise Q2 — Indicate True or False

HF is used for etching glass.

True

HI is a weaker reducing agent as compared to HF.

False

Bleaching powder is completely soluble in water.

False

The formula of perchloric acid is $HClO_2$.

False

On warming, aqueous potassium hypochlorite disproportionates.

True

Alpha-particles emitted by radioactive elements are ions of radon.

False

Radon is the only one of the noble gases that is radioactive.

True

The molecules of the noble gases are all monoatomic.

True

Argon is used to fill electric bulbs.

True

The noble gas which is present in the largest amount in the atmosphere is krypton.

False

## Exercise Q3 — Multiple choice questions

Which of the following hydrogen halides is the weakest acid in solution?

✓ A. HF

B. HBr

C. HI

D. HCl

Chlorine heptaoxide ($Cl_2O_7$) reacts with water to form:

A. Hypochlorous acid

B. Chloric acid

✓ C. Perchloric acid

D. Chlorine and oxygen

Hydrogen bonding is the strongest between the molecules of:

✓ A. HF

B. HCl

C. HBr

D. HI

Which halogen will react spontaneously with Au(s) to produce $Au^{3+}$?

A. Br2

✓ B. F2

C. I2

D. Cl2

The anhydride of HClO4 is:

A. ClO3

B. ClO2

C. Cl2O5

✓ D. Cl2O7

Bleaching powder may be produced by passing chlorine over:

A. calcium carbonate

B. hydrated calcium sulphate

C. anhydrous calcium sulphate

✓ D. calcium hydroxide

Which is the strongest acid?

A. HClO

B. HClO2

C. HClO3

✓ D. HClO4

Which halogen occurs naturally in a positive oxidation state?

A. Fluorine

B. Chlorine

C. Bromine

✓ D. Iodine

An element that has a high ionization energy and tends to be chemically inactive would most likely be:

A. an alkali metal

B. a transition element

✓ C. a noble gas

D. a halogen

Which represents the correct electronic configuration of the outermost energy level of an element of group VIIIA in the ground state?

A. s2p2

B. s2p4

C. s2p5

✓ D. s2p6

**Exercise Q4, Q7, Q11, Q13 — Short questions**

Q4. What is bleaching powder? How is it prepared commercially? Give its uses.

Bleaching powder is calcium chloro-hypochlorite, Ca(OCl)Cl.

It is prepared by passing chlorine gas over dry slaked lime, Ca(OH)2.

It is used for bleaching cotton, linen and paper, and for disinfecting water.

It is also used in the manufacture of chloroform and as an oxidising agent.

Q7(b). Why is HF a weaker acid than HCl?

In water, HF ionises only slightly, while HCl ionises almost completely.

The H-F bond is very strong and short, so it is hard to break.

HF molecules also associate through strong hydrogen bonding.

The small fluoride ion is strongly hydrated, but the bond strength dominates, so HF is weak.

Q11. What are noble gases? Explain their inertness on the basis of their electronic configuration.

Noble gases are the elements of group VIIIA: helium, neon, argon, krypton, xenon and radon.

Except helium (1s2), all have the stable ns2np6 configuration in the outer shell.

Their shells are completely filled, so they have no tendency to gain, lose or share electrons.

They therefore have very high ionization energies and remain chemically inert.

Q13(i). What is iodized salt?

Iodized salt is common table salt mixed with a small amount of potassium iodide or iodate.

It supplies iodine, which the thyroid gland needs to make its hormones.

Lack of iodine causes goitre, a swelling of the thyroid gland.

Iodized salt prevents goitre in people living far from the sea.

Q13(v). Which halogen sublimes to violet vapours?

Iodine sublimes on gentle heating to give violet vapours.

It passes directly from the solid state to the gaseous state without melting.

The vapour condenses back to shining black-violet crystals on cooling.

This property is used to purify iodine by sublimation.

Q13(vii). Which halogen is used in water treatment to kill bacteria?

Chlorine is used to disinfect drinking water and swimming pool water.

It dissolves in water to form hypochlorous acid, a powerful germicide.

Hypochlorous acid kills bacteria and other harmful micro-organisms.

Only a small dose is needed, and it also removes bad odours from water.

**Exercise Q5, Q6, Q7(a), Q8–Q10, Q12, Q13 (continued)**

Q5(a). Discuss the oxides of chlorine.

The oxides of chlorine are: chlorine monoxide, $Cl_2O$; chlorine dioxide, $ClO_2$; dichlorine hexoxide, $Cl_2O_6$; and dichlorine heptoxide, $Cl_2O_7$.

$Cl_2O$ is the anhydride of hypochlorous acid, $ClO_2$ is a powerful oxidising and bleaching agent, and $Cl_2O_7$ is the anhydride of perchloric acid.

All chlorine oxides are strong oxidising agents and are unstable, decomposing

readily to chlorine and oxygen.

Q5(b). What are disproportionation reactions? Explain with an example.

A disproportionation reaction is one in which the same element is both oxidised and reduced in a single reaction.

For example, chlorine with cold dilute sodium hydroxide gives chloride and hypochlorite: Cl2 + 2 NaOH gives NaCl + NaOCl + H2O, where chlorine (0) goes to -1 and +1.

On warming, hypochlorite disproportionates further: 3 ClO- gives 2 Cl- + ClO3-, where chlorine (+1) goes to -1 and +5.

Q6. Discuss the system of nomenclature used for oxyacids of halogens, with examples.

Oxyacids of halogens are named by the oxidation state of the halogen: hypo-...-ous acid for +1, ...-ous acid for +3, ...-ic acid for +5, and per-...-ic acid for +7.

For chlorine: hypochlorous acid, HClO (+1); chlorous acid, HClO2 (+3); chloric acid, HClO3 (+5); perchloric acid, HClO4 (+7).

The salts follow the same pattern: hypochlorite, chlorite, chlorate and perchlorate; acid strength increases with oxidation state.

Q7(a). How are halogen acids ionised in water?

Halogen acids ionise in water by donating a proton to water: HX + H2O gives H3O+ + X-.

Hydrochloric, hydrobromic and hydroiodic acids ionise almost completely, so they are strong acids.

Hydrofluoric acid ionises only slightly, because the H-F bond is very strong and the fluoride ion is heavily hydrated, so HF is a weak acid.

Q8. Arrange in order: (a) increasing acidic character of HClO, HClO2, HClO3, HClO4; (b) increasing oxidising power of F2, Cl2, Br2, I2.

(a) Increasing acidic character: HClO < HClO2 < HClO3 < HClO4, because more

oxygen atoms withdraw electrons from the O-H bond, making it easier to lose H+.

(b) Increasing oxidising power: I2 < Br2 < Cl2 < F2, because oxidising power (electronegativity and reduction potential) decreases down the group, with fluorine the strongest oxidising agent.

Q9. What happens when bleaching powder reacts with: dilute H2SO4, excess concentrated H2SO4, NH3, HI and CO2?

(a) Alkanes are prepared by: hydrogenation of alkenes and alkynes; Wurtz reaction (alkyl halides with sodium); decarboxylation of sodium salts of acids with soda lime; reduction of alkyl halides; and Grignard reagents with water.

(b) (i) Methane to ethane: methane to chloromethane, then Wurtz reaction with sodium gives ethane.

(ii) Ethane to methane: ethane to chloroethane, then to ethanol, then oxidised to acetic acid, then to sodium acetate, then decarboxylation with soda lime gives methane.

(iii) Acetic acid to ethane: acetic acid reduced with LiAlH4 to ethanol, then converted to bromoethane, then reduced with zinc and acid to ethane.

(iv) Methane to nitromethane: methane with concentrated nitric acid at high temperature and pressure gives nitromethane by vapour-phase nitration.

Q10. Discuss the various commercial uses of halogens and their compounds.

Halogens are used as disinfectants and antiseptics (chlorine for water, iodine as tincture), in bleaching (chlorine, bleaching powder), and in making plastics (fluorine in Teflon, chlorine in PVC).

Fluorides are used in toothpaste and water fluoridation; bromides in photography and medicines; iodides in iodized salt and thyroid treatment.

Halogen compounds are also used as solvents, refrigerants (freons), fire extinguishers (carbon tetrachloride, historically), and in dyes and drugs.

Q12. Write notes on: (i) oxyfluorides of xenon, (ii) applications of noble gases.

(i) Oxyfluorides of xenon are compounds in which xenon is bonded to both oxygen and fluorine, for example XeOF2, XeOF4 and XeO2F2.

They are formed by partial hydrolysis of xenon fluorides (for example XeF6 with water gives XeOF4) and are strong oxidising and fluorinating agents.

(ii) Applications of noble gases: helium in balloons and for deep-sea divers' breathing mixtures; neon in advertising signs; argon in electric bulbs and welding; krypton and xenon in lamps and lasers; radon (radioactive) in radiotherapy and, historically, in earthquake prediction studies.

Q13(ii,iii,iv,vi,viii,ix). Short questions: freons and Teflon, ion sizes, iodine lustre, antiseptic halogen, earthquake-prediction gas, bactericidal-lamp gas.

(ii) Freons are chlorofluorocarbons (for example CFCl3, CF2Cl2) used as refrigerants and propellants; Teflon is polytetrafluoroethene (PTFE), a non-stick, heat-resistant polymer made from tetrafluoroethene.

(iii) Increasing ionic size: F- < Cl- < Br- < I-, because each ion has an extra electron shell down the group.

(iv) Iodine has a metallic lustre because its large, easily polarised molecules form shiny, layered crystals that reflect light like a metal surface.

(vi) Iodine is used as an antiseptic, as tincture of iodine (iodine in alcohol) for wounds.

(viii) Radon is the gas used for earthquake prediction, because radon emission from the ground changes before earthquakes.

(ix) Xenon is used in bactericidal lamps, per the chapter's noble-gas uses list.

## Chapter 6: Transition Elements

### Exercise Q1 — Fill in the blanks

The property of paramagnetism is due to the presence of unpaired electrons.

$MnO_4^-$ ion has purple colour and $Cr_2O_7^{2-}$ has orange colour.

When potassium chromate is treated with an acid, potassium dichromate is produced.

The d-block elements are located between s and p block elements.

Oxidation number of Fe in $K_4[Fe(CN)_6]$ is +2 while in $K_3[Fe(CN)_6]$ it is +3.

The presence of moisture in a metal promotes corrosion.

If copper is in contact with aluminium, aluminium gets corroded.

Complexes having $sp^3d^2$ hybridization have octahedral shape.

In naming the complexes, all the ligands are named in alphabetical order .

In an aqueous solution, $CrO_4^{2-}$ and $Cr_2O_7^{2-}$ exist in the form of equilibrium .

## Exercise Q2 — Indicate True or False

A substance which is attracted into a magnetic field is said to be diamagnetic.

False

Compounds of the transition elements are mostly coloured.

True

$Fe^{3+}$ ions are blue when hydrated.

False

An extreme case of paramagnetism is called diamagnetism.

False

Tin plating is used to protect iron sheets from corrosion.

True

In galvanizing, zinc prevents corrosion of iron.

True

Tin plated iron gets rusted more rapidly when the protective coating is damaged than the unplated iron.

True

The name of anionic ligands in a complex ends in suffix 'o'.

True

Pig iron contains greater percentage of carbon than steel.

True

Complex compounds having dsp2 hybridization have tetrahedral geometry.

False

## Exercise Q3 — Multiple choice questions

Which of the following is a non-typical transition element?

A. Cr

B. Mn

✓ C. Zn

D. Fe

Which of the following is a typical transition metal?

A. Sc

B. Y

C. Ra

✓ D. Co

f-block elements are also called:

A. non-typical transition elements

✓ B. outer transition elements

C. normal transition elements

D. None is true

The strength of binding energy of transition elements depends upon:

A. number of electron pairs

✓ B. number of unpaired electrons

C. number of neutrons

D. number of protons

Group VIB of transition elements contains:

A. Zn, Cd, Hg

B. Fe, Ru, Os

✓ C. Cr, Mo, W

D. Mn, Te, Re

Which is the formula of tetraammine chloro-nitro-platinum (IV) sulphate?

A. [Pt(NH3)4(NO2)]SO4

B. [Pt NO2 2Cl(NH3)4]SO4

C. [Pt Cl(NO2)(NH3)4]SO4

✓ D. [Pt(NH3)4(NO2)Cl]SO4

The percentage of carbon in different types of iron products is in the order of:

A. cast iron > wrought iron > steel

B. wrought iron > steel > cast iron

✓ C. cast iron > steel > wrought iron

D. cast iron = steel > wrought iron

The colour of transition metal complexes is due to:

✓ A. d-d transition of electrons

B. paramagnetic nature of transition elements

C. ionization

D. loss of s-electrons

Coordination number of Pt in [Pt Cl(NO2)(NH3)4] is:

A. 2

B. 4

C. 1

✓ D. 6

The total number of transition elements is:

A. 10

B. 14

✓ C. 40

D. 58

**Exercise Q4, Q5, Q8 — Short questions**

Q4. How does the electronic configuration of the valence shell affect: (a) binding energy (b) paramagnetism (c) melting points (d) oxidation states?

(a) More unpaired d-electrons give stronger metallic bonding, so binding energy rises to a peak in the middle of a series.

(b) Unpaired electrons make the metal paramagnetic; the moment grows with the number of unpaired electrons.

(c) Stronger bonding from unpaired electrons also raises melting and boiling points toward the middle.

(d) The ns and (n-1)d electrons can both take part in bonding, giving many variable oxidation states.

Q5. Explain: (a) Ligands (b) Coordination sphere (c) Substitutional alloy (d) Central metal atom.

(a) A ligand is an ion or molecule that donates an electron pair to a metal, like NH3, Cl- or CN-.

(b) The coordination sphere is the central metal plus its attached ligands, written in square brackets.

(c) A substitutional alloy forms when atoms of one metal replace atoms of another in the crystal, like brass.

(d) The central metal atom is the metal ion that accepts electron pairs from the ligands in a complex.

Q8(c). How does the process of galvanizing protect iron from rusting?

In galvanizing, iron is coated with a thin layer of zinc.

Zinc is more reactive than iron, so it acts as a sacrificial anode.

Even if the coating is scratched, zinc corrodes first and iron stays protected.

Zinc also forms a basic carbonate layer which seals the surface from air and moisture.

**Exercise Q6, Q7, Q8(a,b), Q9–Q11 (continued)**

Q6. Describe the rules for naming coordination complexes and give examples.

Rules for naming coordination complexes: the cation is named before the anion; ligands are named before the central metal, in alphabetical order ignoring numerical prefixes.

Anionic ligands end in -o (chloro, cyano, hydroxo), neutral ligands keep their names except aqua (water), ammine (ammonia) and carbonyl.

The oxidation state of the central metal is shown in Roman numerals in brackets, and for anionic complexes the metal name ends in -ate (for example hexacyanoferrate).

Examples: [Co(NH3)6]Cl3 is hexaamminecobalt(III) chloride; K4[Fe(CN)6] is potassium hexacyanoferrate(II).

Q7. What is the difference between wrought iron and steel? Explain Bessemer's process.

Wrought iron is almost pure iron (0.12-0.25% carbon), soft and malleable, while steel contains 0.25-2.25% carbon and is harder and stronger.

Bessemer's process makes steel from pig iron: molten pig iron is placed in a Bessemer converter and a blast of air is blown through it.

The air oxidises silicon, manganese and carbon impurities, and the heat of oxidation keeps the metal molten; measured amounts of spiegeleisen

(iron-manganese-carbon alloy) are then added to give steel of the required carbon content.

Q8(a,b). (a) Why damaged tin-plated iron rusts quickly. (b) Under what conditions aluminium corrodes.

(a) Damaged tin-plated iron rusts quickly because tin is less reactive than iron: when the coating breaks, iron becomes the anode of a galvanic cell and corrodes faster than unplated iron, while tin acts as the cathode.

(b) Aluminium normally resists corrosion because of a protective oxide layer, but it corrodes when that layer is removed by acids, alkalies, chloride solutions (sea water), or contact with a more noble metal, or by mercury salts.

Q9. How are chromate ions converted into dichromate ions?

Chromate ions are converted to dichromate ions by acid: 2 CrO4(2-) + 2 H+ gives Cr2O7(2-) + H2O.

The equilibrium is reversed by alkali, which converts dichromate back to chromate. The change is visible: the solution changes from yellow (chromate) to orange (dichromate).

Q10. Describe the preparation of KMnO4 and K2CrO4.

Potassium permanganate is prepared by fusing manganese dioxide with potassium hydroxide in air to give potassium manganate, K2MnO4, which is then oxidised electrolytically (or with chlorine) to KMnO4.

Potassium chromate is prepared by roasting chromite ore (FeCr2O4) with sodium carbonate in air to give sodium chromate, which is converted to potassium chromate by adding potassium chloride.

Both preparations use oxidation of the lower oxidation state to the higher one, followed by crystallisation.

Q11. Give systematic names to the complexes (a)-(h).

(a) [Fe(CO)5] is pentacarbonyliron(0). (b) [Co(NH3)6]Cl3 is hexaamminecobalt(III)

chloride. (c) [Fe(H2O)6]2+ is hexaaquairon(II) ion.

(d) Na3[CoF6] is sodium hexafluorocobaltate(III). (e) K2[Cu(CN)4] is potassium tetracyanocuprate(II). (f) K2[PtCl6] is potassium hexachloroplatinate(IV). (g) [Pt(OH)2(NH3)4]SO4 is tetraamminedihydroxidoplatinum(IV) sulphate. (h) [Cr(OH)3(H2O)3] is trihydroxidotriaquachromium(III).

## Chapter 7: Fundamental Principles of Organic Chemistry

### Exercise Q1 — Fill in the blanks

Organic compounds having the same molecular formula but different structural formulas are called isomers.

The state of hybridization of the carbon atom in ethene is sp2 .

Alkenes show cis-trans isomerism due to restricted rotation around a carbon-carbon double bond.

Heating an organic compound in the absence of oxygen and in the presence of aluminium silicate as a catalyst is called cracking.

A group of atoms which confers characteristic properties on an organic compound is called a functional group .

2-Butene is a position isomer of 1-butene.

Carbonyl functional group is present in both aldehydes and ketones.

A heterocyclic compound contains an atom other than carbon in its ring.

The quality of gasoline can be checked by finding out its octane number .

A carboxylic acid contains carboxyl group (-COOH) as a functional group.

### Exercise Q2 — Indicate True or False

There are three possible isomers for pentane.

True

Alkynes do not show the phenomenon of cis-trans isomerism.

True

Organic compounds cannot be synthesized from inorganic compounds.

False

All closed chain compounds are aromatic in nature.

False

The functional group present in amides is called an amino group.

False

Government of Pakistan is trying to use coal for power generation.

True

Crude petroleum is subjected to fractional sublimation in order to separate it into different fractions.

False

A bond between carbon and hydrogen serves as a functional group for alkanes.

True

o-Nitrotoluene and p-nitrotoluene are examples of functional group isomerism.

False

Almost all the chemical reactions taking place in our body are inorganic in nature.

False

## Exercise Q3 — Multiple choice questions

The state of hybridization of the carbon atom in methane is:

✓ A. sp3

B. sp2

C. sp

D. dsp2

In t-butyl alcohol, the tertiary carbon is bonded to:

A. two hydrogen atoms

B. three hydrogen atoms

C. one hydrogen atom

✓ D. no hydrogen atom

Which set of hybrid orbitals has planar triangular shape?

A. sp3

B. sp

✓ C. sp2

D. dsp2

The chemist who synthesized urea from ammonium cyanate was:

A. Berzelius

B. Kolbe

✓ C. Wohler

D. Lavoisier

Linear shape is associated with which set of hybrid orbitals?

✓ A. sp

B. sp2

C. sp3

D. dsp2

A double bond consists of:

A. two sigma bonds

✓ B. one sigma and one pi bond

C. one sigma and two pi bonds

D. two pi bonds

Ethers show the phenomenon of:

A. position isomerism

B. functional group isomerism

✓ C. metamerism

D. cis-trans isomerism

Select from the following the one which is an alcohol:

✓ A. CH3-CH2-OH

B. CH3-O-CH3

C. CH3COOH

D. CH3-CH2-Br

**Exercise Q4, Q8, Q9, Q10, Q13 — Short questions**

Q4. How are organic compounds classified? Give a suitable example of each type.

Organic compounds are first divided into acyclic (open chain) and cyclic (closed chain) compounds.

Acyclic example: ethane, CH3-CH3, with an open chain of carbon atoms.

Cyclic compounds are homocyclic, with only carbon in the ring, like benzene.

They are also heterocyclic, with another atom like nitrogen or oxygen in the ring, like pyridine.

Q8. What is meant by a functional group? Name typical functional groups containing oxygen.

A functional group is an atom or group of atoms that gives an organic compound its characteristic properties.

Compounds with the same functional group show similar chemical reactions.

Oxygen-containing groups are: hydroxyl (-OH) in alcohols, alkoxy (-OR) in ethers.

Carbonyl (>C=O) in aldehydes and ketones, and carboxyl (-COOH) in carboxylic acids.

Q9. What is an organic compound? Explain the importance of Wohler's work.

An organic compound is a compound of carbon, usually with hydrogen, oxygen or

nitrogen.

Earlier chemists believed organic compounds could only be made by living things, the vital force theory.

In 1828 Wohler heated ammonium cyanate, an inorganic salt, and obtained urea, an organic compound.

This destroyed the vital force theory and opened the laboratory synthesis of organic compounds.

Q10. Write a short note on cracking of hydrocarbons.

Cracking is the breaking of long-chain hydrocarbons into smaller, more useful molecules.

It is done by heating heavy fractions strongly in the absence of oxygen, with a catalyst.

For example, decane cracks to give octane and ethene.

Cracking increases the yield of petrol and provides alkenes for the chemical industry.

Q13. What is orbital hybridization? Explain sp3, sp2 and sp modes of hybridization of carbon.

Hybridization is the mixing of atomic orbitals of similar energy to form new hybrid orbitals.

In sp3 hybridization, one s and three p orbitals mix to give four tetrahedral orbitals, as in methane.

In sp2 hybridization, one s and two p orbitals give three planar triangular orbitals, as in ethene.

In sp hybridization, one s and one p orbital give two linear orbitals, as in ethyne.

**Exercise Q5–Q7, Q11, Q12, Q14, Q15 (continued)**

Q5. What are homocyclic and heterocyclic compounds? Give one example of each.

Homocyclic compounds have a ring made of carbon atoms only, for example benzene and cyclohexane.

Heterocyclic compounds have a ring containing at least one atom other than carbon, for example pyridine (nitrogen in the ring), furan (oxygen) and thiophene (sulphur).

Heterocyclic compounds are very important in biology: the bases of DNA and many drugs are heterocyclic.

Q6. Write the structural formulas of the two possible isomers of C4H10.

The two isomers of C4H10 are n-butane (CH3-CH2-CH2-CH3, a straight chain) and 2-methylpropane (CH3-CH(CH3)-CH3, a branched chain).

Both have the same molecular formula but different structures and properties: n-butane boils at about 0 C, while 2-methylpropane boils at about -12 C.

They are chain isomers, the simplest example of structural isomerism in alkanes.

Q7. Why is ethene an important industrial chemical?

Ethene is an important industrial chemical because it is the raw material for many products.

It is polymerised to polythene (plastics, bags, bottles), hydrated to ethyl alcohol, and converted to ethylene glycol (antifreeze, polyester) and to chloroethene (PVC).

It is also used to ripen fruits artificially and to make ethylene oxide, acetaldehyde and many other chemicals.

Q11. Explain reforming of petroleum with a suitable example.

Reforming is the conversion of straight-chain hydrocarbons of low octane number into branched-chain and aromatic hydrocarbons of high octane number, using heat and a catalyst (platinum).

For example, n-heptane (octane number 0) is reformed to toluene, an aromatic compound with a high octane number, by cyclisation and dehydrogenation.

Reforming raises the quality of petrol and reduces engine knocking.

Q12. Describe important sources of organic compounds.

The important sources of organic compounds are coal, petroleum and natural gas, which supply most hydrocarbons and their derivatives.

Plants and animals supply sugars, starch, cellulose, fats, proteins and natural rubber.

Many organic compounds are now also made synthetically in the laboratory and industry, for example urea, dyes, plastics and drugs, from simple starting materials.

Q14. Explain the type of bonds and shapes of CH3-CH3, CH2=CH2, CH triple CH, HCHO and CH3Cl using hybridisation.

Ethane: each carbon is sp3 hybridised, forming four sigma bonds in a tetrahedral shape (bond angle about 109.5 degrees).

Ethene: each carbon is sp2 hybridised, forming three sigma bonds in a plane (about 120 degrees) plus one pi bond, giving a planar molecule.

Ethyne: each carbon is sp hybridised, forming two sigma bonds in a line (180 degrees) plus two pi bonds, giving a linear molecule.

Formaldehyde (HCHO): carbon is sp2 hybridised, trigonal planar. Chloromethane (CH3Cl): carbon is sp3 hybridised, tetrahedral.

Q15. Why is there no free rotation around a double bond and free rotation around a single bond? Discuss cis-trans isomerism.

There is free rotation around a single (sigma) bond because the sigma bond is cylindrically symmetrical about the bond axis, so rotation does not break it.

There is no free rotation around a double bond because rotation would break the pi bond, which needs the p orbitals to stay parallel; breaking it costs much energy.

This restricted rotation gives cis-trans isomerism in alkenes: when the same groups are on the same side of the double bond it is cis, and when on opposite sides it is trans, provided each double-bond carbon carries two different groups.

# Chapter 8: Aliphatic Hydrocarbons

## Exercise Q1 — Fill in the blanks

Ozone reacts with ethene to form formaldehyde .

Lindlar's catalyst is used for partial hydrogenation of alkynes.

Divinyl acetylene is a conjugated acetylene.

Vicinal dihalides have two halogens on adjacent carbon atoms.

Ethyne is acidic in character because of sp hybridization.

Halohydrins are formed due to addition of hypohalous acid in ethene.

Ethylene glycol is produced when ethene reacts with cold alkaline KMnO4 solution.

Mustard gas is a high boiling liquid .

Ethyne has garlic-like odour.

Ethyne is obtained by the reaction of water with calcium carbide.

## Exercise Q2 — Indicate True or False

Addition of HX to unsymmetrical alkenes takes place according to Markownikov's rule.

True

Methane reacts with bromine water and its colour is discharged.

False

Mustard gas is a blistering agent.

True

Methane is also called marsh gas.

True

Ethyne is a saturated compound.

False

Baeyer's reagent is used to locate a double bond in an alkene.

True

Alkanes usually undergo substitution reactions.

True

Benzene is a polymer of ethene.

False

Acrylonitrile can be obtained from ethyne.

True

Ethyne is more reactive towards electrophilic reagents than ethene.

False

## Exercise Q3 — Multiple choice questions

Preparation of vegetable ghee involves:

A. Halogenation

✓ B. Hydrogenation

C. Hydroxylation

D. Dehydrogenation

Formula of chloroform is:

A. $CH_3Cl$

B. $CCl_4$

C. $CH_2Cl_2$

✓ D. $CHCl_3$

The presence of a double bond in a compound is the sign of:

A. Saturation

✓ B. Unsaturation

C. Substitution

D. None

Vinyl acetylene combines with HCl to form:

A. Polyacetylene

B. Benzene

✓ C. Chloroprene

D. Divinyl acetylene

The addition of an unsymmetrical reagent to an unsymmetrical alkene is in accordance with the rule of:

A. Hund's rule

✓ B. Markownikov's rule

C. Pauli's Exclusion Principle

D. Aufbau Principle

Synthetic rubber is made by polymerization of:

A. Chloroform

B. Acetylene

C. Divinylacetylene

✓ D. Chloroprene

Beta-beta-dichloroethyl sulphide is commonly known as:

✓ A. Mustard gas

B. Laughing gas

C. Phosgene gas

D. Bio-gas

When methane reacts with Cl2 in the presence of diffused sunlight, the products obtained are:

A. Chloroform only

B. Carbon tetrachloride only

C. Chloromethane and dichloromethane

✓ D. Mixture of chloromethanes

Which one of the following gases is used for artificial ripening of fruits?

✓ A. Ethene

B. Ethyne

C. Methane

D. Propane

**Exercise Q6, Q8 — Short questions**

Q6. What are the rules for naming alkanes? Explain with suitable examples.

First, select the longest continuous carbon chain as the parent chain.

Second, number the chain so that the substituents get the lowest possible numbers.

Third, name substituents alphabetically, with their position numbers, like 2-methylbutane.

If the same substituent occurs twice, use di, tri, as in 2,2-dimethylpropane.

Q8(a). Explain why alkanes are less reactive than alkenes. What is the effect of branching on the melting point of alkanes?

Alkanes contain only strong C-C and C-H sigma bonds, which are hard to break.

Alkenes contain a pi bond, which is weaker and easily attacked by electrophiles.

So alkanes are less reactive and undergo substitution only under special conditions.

Branching makes molecules more compact, so they pack better and their melting points rise.

**Exercise Q4, Q5, Q7–Q14, Q16–Q26 (continued)**

Q4. Write the structural formula (in words) for each compound (i)-(viii).

In words: (i) 2-methylpropane is CH3-CH(CH3)-CH3. (ii) Neopentane is the same as 2,2-dimethylpropane, C(CH3)4.

(iii) 3-Ethylpentane is CH3-CH2-CH(C2H5)-CH2-CH3. (iv) 4-Ethyl-3,4-dimethylheptane is a 7-carbon chain with an ethyl group at carbon 4 and methyl groups at carbons 3 and 4.

(v) 2,2,3,4-Tetramethylpentane is a 5-carbon chain with two methyls at carbon 2 and one each at carbons 3 and 4.

(vi) 4-Isopropylheptane is a 7-carbon chain with an isopropyl group at carbon 4. (vii) 2,2-Dimethylbutane is CH3-C(CH3)2-CH2-CH3. (viii) 2,2-Dimethylpropane is C(CH3)4.

Q5. Names by IUPAC system.

Text-given items: (ii) (CH3)3C-CH2-C(CH3)3 is 2,2,4,4-tetramethylpentane. (v) CH3CH2C(CH3)2CH(CH2CH3)CH3 is 3,3-dimethylhexane.

(vi) (CH3CH2)3CH is 3-ethylpentane. (vii) CH3C(CH3)2(CH2)2CH3 is 2,2-dimethylpentane. (viii) (C6H5)3CH is triphenylmethane.

Diagram items, read from book p.45: (i) CH3-CH2-CH(C2H5)-CH2-CH3 is 3-ethylpentane. (iii) CH3-CH(CH3)-CH2-CH(CH3)2 is 2,4-dimethylpentane. (iv) (CH3)2CH-CH(CH3)-CH(CH3)2 is 2,3,4-trimethylpentane.

Q7. (a) Structural formulas (in words) for all isomeric hexanes. (b) Correct the incorrect IUPAC names.

The isomeric hexanes are: n-hexane (CH3CH2CH2CH2CH2CH3); 2-methylpentane; 3-methylpentane; 2,2-dimethylbutane; and 2,3-dimethylbutane.

(b) Correct names: (i) 4-methylpentane is incorrect because the longest chain is 5 carbons numbered to give the methyl the lowest locant: it is 2-methylpentane.

(ii) 3,5,5-Trimethylhexane is incorrect: the longest chain is 7 carbons, giving 2,2,4-trimethylheptane. (iii) 2-Methyl-3-ethylbutane is incorrect: the longest chain is 5 carbons, giving 3-methylpentane.

Q8(b). Three different unsaturated compounds yield 2-methylbutane when they are hydrogenated. Give their structures and write equations for the reactions involved.

(b) Three different alkenes give 2-methylbutane on hydrogenation:

2-methyl-1-butene, 2-methyl-2-butene and 3-methyl-1-butene.

Each adds one molecule of hydrogen across its double bond over a metal catalyst to

give the same saturated product, 2-methylbutane.

This shows that hydrogenation removes the double bond but keeps the carbon skeleton, so different double-bond positions give the same alkane.

Q9. (a) Methods of preparation of alkanes. (b) Conversions: methane to ethane, ethane to methane, acetic acid to ethane, methane to nitromethane.

(a) Alkanes are prepared by: hydrogenation of alkenes and alkynes; Wurtz reaction (alkyl halides with sodium); decarboxylation of sodium salts of acids with soda lime; reduction of alkyl halides; and Grignard reagents with water.

(b) (i) Methane to ethane: methane to chloromethane, then Wurtz reaction with sodium gives ethane.

(ii) Ethane to methane: ethane to chloroethane, then to ethanol, then oxidised to acetic acid, then to sodium acetate, then decarboxylation with soda lime gives methane.

(iii) Acetic acid to ethane: acetic acid reduced with $LiAlH_4$ to ethanol, then converted to bromoethane, then reduced with zinc and acid to ethane.

(iv) Methane to nitromethane: methane with concentrated nitric acid at high temperature and pressure gives nitromethane by vapour-phase nitration.

Q10. (a) Octane number and knocking. (b) Free radical mechanism of chlorine with methane in sunlight.

(a) Octane number is a measure of a fuel's resistance to knocking (premature explosion) in an engine, rated against iso-octane (100) and n-heptane (0).

A high-octane fuel burns smoothly and evenly, so it has less tendency to knock, because it resists auto-ignition under compression.

(b) Free radical mechanism for chlorine with methane in sunlight: initiation: $Cl_2$ splits into two chlorine free radicals by sunlight.

Propagation: a chlorine radical removes hydrogen from methane to give methyl radical and HCl; the methyl radical reacts with $Cl_2$ to give chloromethane and a new

chlorine radical.

Termination: two radicals combine, for example two methyl radicals give ethane, or a methyl and a chlorine radical give chloromethane, ending the chain.

Q11. (a) Structural formulas (in words) for (i)-(xii). (b) IUPAC names.

(a) In words: (i) isobutylene is (CH3)2C=CH2 (2-methylpropene). (ii) 2,3,4,4-Tetramethyl-2-pentene is a 5-carbon chain with a double bond at carbon 2 and methyls at 2, 3, 4 and 4.

(iii) 2,5-Heptadiene is a 7-carbon chain with double bonds at carbons 2 and 5. (iv) 4,5-Dimethyl-2-hexene is a 6-carbon chain with a double bond at carbon 2 and methyls at 4 and 5.

(v) Vinylacetylene is CH2=CH-C triple CH (but-1-en-3-yne). (vi) 1,3-Pentadiene is CH2=CH-CH=CH-CH3.

(vii) 1-Butyne is CH triple C-CH2-CH3. (viii) 3-n-Propyl-1,4-pentadiene is a 5-carbon chain with double bonds at 1 and 4 and a propyl group at carbon 3.

(ix) Vinyl bromide is CH2=CH-Br. (x) But-1-en-3-yne is CH2=CH-C triple CH. (xi) 4-Methyl-2-pentyne is CH3-C triple C-CH(CH3)-CH3 with the triple bond at carbon 2. (xii) Isopentane is 2-methylbutane.

(b) Text-given items: (i) CH3-CH=CH-(CH2)2CH3 is hex-2-ene. (ii) (CH3)2C=CH2 is 2-methylpropene.

(iii) The compound with a propyl chain and an isopropyl group on the double-bond carbon is 2-isopropylpent-1-ene. (v) CH2=C(C2H5)CH2CH2CH3 is 2-ethylpent-1-ene. (vi) CH triple C-CH3 is propyne. (ix) CH triple C-CH=CH-C triple CH is hex-3-en-1,5-diyne. (x) CH2=CH-C triple CH is but-1-en-3-yne.

Diagram items, read from book p.46: (iv) CH2=CH-CH=CH2 is buta-1,3-diene. (vii) CH3-C triple C-CH3 is but-2-yne. (viii) CH2=CH-C triple C-CH=CH2 is hexa-1,5-dien-3-yne.

Q12. (a) Preparation of alkenes; proof of the double bond in ethylene. (b)

Dehydrohalogenation products.

(a) Alkenes are prepared by: dehydration of alcohols with concentrated sulphuric acid; dehydrohalogenation of alkyl halides with alcoholic KOH; hydrogenation of alkynes with Lindlar's catalyst; and cracking of alkanes.

Ethylene contains a double bond, proved because it decolourises bromine water (addition) and cold dilute alkaline KMnO4, Baeyer's reagent (hydroxylation to a diol), while ethane does not.

(b) Dehydrohalogenation with a strong base: (i) 1-chloropentane gives pent-1-ene.

(ii) 2-Chloro-3-methylbutane gives mainly 2-methylbut-2-ene (Saytzeff product) and some 2-methylbut-1-ene.

(iii) 1-Chloro-2,2-dimethylpropane (neopentyl chloride) gives 2-methylbut-2-ene by rearrangement, because the primary carbocation rearranges to the more stable tertiary carbocation before losing a proton.

Q13. (a) Preparation of propene. (b) All possible alkenes of formula C5H10 (in words).

(a) Propene from: (i) propan-1-ol by dehydration with concentrated H2SO4; (ii) propyne by partial hydrogenation with Lindlar's catalyst; (iii) isopropyl chloride by dehydrohalogenation with alcoholic KOH.

(b) Skeleton formulas in words for C5H10 alkenes: pent-1-ene, pent-2-ene (cis and trans), 2-methylbut-1-ene, 2-methylbut-2-ene, 3-methylbut-1-ene, and cyclopentane (a cycloalkane isomer).

Q14. (a) Ethene to ethyl alcohol. (b) From ethene: 1,2-dibromoethane, ethyne, ethane, ethylene glycol. (c) Conversions.

(a) Ethene is converted to ethyl alcohol by hydration with steam over a phosphoric acid catalyst at high temperature and pressure, or via ethyl hydrogen sulphate with concentrated sulphuric acid followed by hydrolysis.

(b) From ethene: (i) with bromine, 1,2-dibromoethane; (ii) 1,2-dibromoethane with

alcoholic KOH gives ethyne; (iii) ethene with hydrogen over nickel gives ethane; (iv) ethene with cold dilute alkaline KMnO4 gives ethylene glycol.

(c) (i) 1-Butene to 1-butyne: 1-butene with bromine gives 1,2-dibromobutane, then with alcoholic KOH (twice) gives but-1-yne.

(ii) 1-Propanol to the chloro compound printed as CH3-CH-CH2Cl: 1-propanol with PCl5 or SOCl2 gives 1-chloropropane, CH3-CH2-CH2Cl (the printed line compresses the middle CH2; no branch substituent is shown on the render).

Q16. Structural formulas (in words) for the products of 1-butene with the reagents listed.

(i) With H2/Pt, 1-butene gives butane. (ii) With Br2 in CCl4, it gives 1,2-dibromobutane. (iii) With cold dilute KMnO4/OH-, it gives butane-1,2-diol. (iv) With HBr, it gives mainly 2-bromobutane (Markovnikov addition). (v) With O2 in the presence of silver, it gives 1,2-epoxybutane.

(vi) With HOCl, it gives 2-chlorobutan-1-ol (chlorohydrin, OH on carbon 1, Cl on carbon 2 by Markovnikov orientation of Cl+). (vii) With dilute H2SO4, it gives mainly butan-2-ol.

Q17. Identify each lettered product (A-F).

(i) Ethyl alcohol with concentrated H2SO4 gives ethene (A); ethene with bromine gives 1,2-dibromoethane (B); B with alcoholic KOH gives ethyne (C).

(ii) Propene with bromine gives 1,2-dibromopropane (D); D with alcoholic KOH gives propyne (E); E with HCN gives 2-cyanopropene (F), by Markovnikov addition of HCN across the triple bond.

Q18. After ozonolysis the only product was acetaldehyde. Guess the structural formula.

Ozonolysis of an alkene breaks the double bond and gives carbonyl compounds from each side.

If the only product is acetaldehyde, CH3CHO, then both sides of the double bond

gave acetaldehyde, so the alkene was symmetric with CH3-CH= on both sides.

The compound is therefore but-2-ene, CH3-CH=CH-CH3.

Q19. (a) Alcohols from addition of sulphuric acid to propene, 1-butene, 2-butene. (b) Product of HCl addition to 2-methyl-2-butene.

(a) Addition of sulphuric acid follows Markovnikov's rule (H+ adds to the carbon with more hydrogens, giving the more stable carbocation, then HSO4- adds).

(i) Propene gives propan-2-ol (after hydrolysis of isopropyl hydrogen sulphate). (ii) 1-Butene gives butan-2-ol. (iii) 2-Butene gives butan-2-ol.

(b) Addition of HCl to 2-methyl-2-butene gives 2-chloro-2-methylbutane: H+ adds to give the tertiary carbocation at carbon 2, which is attacked by Cl-, because the tertiary carbocation is the most stable.

Q20. Why are some hydrocarbons called saturated and others unsaturated? What reactions are characteristic of them?

Hydrocarbons are called saturated when all carbon-carbon bonds are single bonds and each carbon carries the maximum number of hydrogens (alkanes).

Unsaturated hydrocarbons contain double or triple bonds (alkenes, alkynes) and have fewer hydrogens than the corresponding alkane.

Saturated hydrocarbons characteristically undergo substitution reactions, while unsaturated hydrocarbons characteristically undergo addition reactions across the multiple bond.

Q21(first set). (a) Preparation of ethyne. (b) Reactions of ethyne. (c) Uses of methane, ethene and ethyne.

(a) Ethyne is prepared by: the reaction of calcium carbide with water, CaC2 + 2 H2O gives C2H2 + Ca(OH)2; and by cracking of methane or other hydrocarbons at high temperature.

(b) Ethyne reacts: (i) with hydrogen over a catalyst to give ethene and then ethane; (ii) with halogen acids to give vinyl halide and then gem-dihalide (Markovnikov).

(iii) With alkaline KMnO4 it is oxidised (the purple colour is discharged, giving oxalic acid and then carbon dioxide). (iv) With 10% H2SO4 and HgSO4 it hydrates to acetaldehyde.

(v) With ammoniacal cuprous chloride it gives a red precipitate of copper acetylide, Cu2C2, proving its terminal alkyne character.

(c) Uses: methane as a fuel and for making hydrogen, carbon black and chloroform; ethene for polythene, ethanol, ethylene glycol and fruit ripening; ethyne for oxy-acetylene welding and cutting, and for making vinyl chloride, acetaldehyde and acetic acid.

Q22. Distinguish ethane, ethene and ethyne by chemical reactions.

Ethane gives no reaction with bromine water or with ammoniacal cuprous chloride, and does not discharge KMnO4.

Ethene decolourises bromine water and alkaline KMnO4, but gives no precipitate with ammoniacal cuprous chloride or silver nitrate.

Ethyne decolourises bromine water and KMnO4, and gives a red precipitate with ammoniacal cuprous chloride and a white precipitate with ammoniacal silver nitrate, because it is a terminal alkyne.

Q23. Synthesise from ethyne: acetaldehyde, benzene, chloroprene, glyoxal, oxalic acid, acrylonitrile, ethane, methyl nitrile. (b) Acidity of ethyne.

From ethyne: (i) acetaldehyde by hydration with H2SO4/HgSO4; (ii) benzene by cyclic polymerisation over red-hot iron tube (3 C2H2 gives C6H6).

(iii) Chloroprene by addition of HCl to vinylacetylene (itself from ethyne dimerisation); (iv) glyoxal by oxidation of ethyne with nitric acid or by ozonolysis routes.

(v) Oxalic acid by oxidation of ethyne; (vi) acrylonitrile by addition of HCN to ethyne; (vii) ethane by complete hydrogenation; (viii) methyl nitrile (acetonitrile) by addition of HCN followed by steps, or from ethyne via acetaldehyde routes in the book's

scheme.

(b) Ethyne is acidic because its carbon is sp hybridised with 50% s character, so the C-H bond electrons are held close to carbon and the proton is easily lost.

Terminal alkynes therefore react with ammoniacal silver nitrate and cuprous chloride to give metal acetylides, while internal alkynes do not.

Q24. (a) Compare reactivity of ethane, ethene and ethyne. (b) Compare physical properties of alkanes, alkenes and alkynes.

(a) Ethane, ethene and ethyne differ in reactivity: ethyne is the most reactive toward electrophilic addition in some contexts, but toward catalytic hydrogenation the order is ethyne > ethene > ethane, because multiple bonds are more reactive than single bonds.

In practice, ethene is more reactive than ethyne toward electrophiles because the pi electrons of ethyne are held more tightly (sp hybridisation).

(b) Physical properties: alkanes are non-polar, insoluble in water, with boiling points rising with chain length; alkenes are similar but slightly more reactive and with similar boiling points; alkynes have slightly higher boiling points than corresponding alkenes and are also non-polar and water-insoluble.

Q25. How does propyne react with: $AgNO_3/NH_4OH$, $Cu_2Cl_2/NH_4OH$, $H_2O/H_2SO_4/HgSO_4$?

Propyne reacts: (a) with ammoniacal silver nitrate to give a white precipitate of silver propynide (silver acetylide), because it is a terminal alkyne.

(b) With ammoniacal cuprous chloride to give a red precipitate of copper propynide.

(c) With water, $H_2SO_4$ and $HgSO_4$ (hydration, Markovnikov) to give acetone (propanone) via the enol intermediate.

Q26. A compound $C_4H_6$, hydrogenated to $C_4H_{10}$, gives a white precipitate with ammoniacal silver nitrate. Identify it.

The compound $C_4H_6$ is hydrogenated to $C_4H_{10}$ (butane), so it has two degrees of

unsaturation (two double bonds, or one triple bond, or a ring plus a double bond).

It gives a white precipitate with ammoniacal silver nitrate, so it is a terminal alkyne. The compound is therefore but-1-yne, CH triple C-CH2-CH3, which hydrogenates to butane and forms silver acetylide.

Q21(second set, book p. 49). (a) Identify A and B. (b) General mechanism of electrophilic addition to alkenes.

(a) Propan-1-ol with PCl5 gives 1-chloropropane (A); A with sodium in ether (Wurtz reaction) gives hexane (B): 2 CH3CH2CH2Cl + 2 Na gives CH3(CH2)4CH3 + 2 NaCl.

(b) General mechanism of electrophilic addition to alkenes: the pi bond attacks the electrophile (for example H+), forming the more stable carbocation intermediate. The nucleophile (for example Br- or Cl-) then attacks the carbocation, giving the addition product; the orientation follows Markovnikov's rule for unsymmetrical alkenes.

Q15. Cycle of changes: ethane, ethene, ethyne. Chapter 8 numbering note.

Cycle, in words: Ethane gives ethene by dehydrogenation (heating with a catalyst). Ethene gives ethyne in two steps: ethene with bromine gives 1,2-dibromoethane, and 1,2-dibromoethane with alcoholic KOH (twice) gives ethyne.

Ethyne gives ethane back by hydrogenation with hydrogen over a nickel catalyst (ethyne + 2H2 gives ethane), closing the cycle.

Numbering note (render-verified): Q3(viii) is not printed — book p.44 jumps from (vii) to (ix); this is a book numbering skip, not an omitted question.

## Chapter 9: Aromatic Hydrocarbons

### Exercise Q1 — Fill in the blanks

The term aromatic was derived from the Greek word aroma meaning pleasant

smell.

Aromatic hydrocarbons include benzene and all those compounds which are structurally related to benzene.

Benzene is recognized as the simplest member of the class of aromatic hydrocarbons.

Benzene has a cyclic planar hexagonal structure.

The removal of a hydrogen atom from aromatic hydrocarbons gives a radical. The radicals are called aryl radicals .

Benzene was discovered by Michael Faraday in 1825 .

The unhybridized 2pz orbitals in benzene partially overlap to form a pi-cloud of electron cloud.

The introduction of a halogen group in the benzene ring is called halogenation .

## Exercise Q3 — Multiple choice questions (selection)

Aromatic hydrocarbons are the derivatives of:

A. normal series of paraffins

B. alkene

✓ C. benzene

D. cyclohexane

Which of the following can be used as a catalyst in Friedel-Crafts reactions?

✓ A. AlCl3

B. HNO3

C. BeCl2

D. NaCl

## Exercise Q4–Q6 — Short questions

Q4. What are aromatic hydrocarbons? How are they classified?

Aromatic hydrocarbons are benzene and compounds structurally related to benzene.

They contain a benzene ring with delocalised pi electrons, which makes them unusually stable.

They are classified as monocyclic, like benzene and toluene, and polycyclic, like naphthalene.

They may also be benzenoid, containing benzene rings, or non-benzenoid.

Q5(a). What happens when benzene is heated with conc. H2SO4 at 250 C?

At ordinary temperature, benzene gives benzene sulphonic acid by electrophilic substitution.

At 250 C with concentrated sulphuric acid, a second sulphonic acid group enters the ring.

The product is benzene-1,3-disulphonic acid.

The reaction is a sulphonation, and the sulpho group can later be replaced by other groups.

Q5(b). What happens when chlorine is passed through benzene in sunlight?

In sunlight, benzene undergoes free radical addition with chlorine.

Three molecules of chlorine add across the ring to give benzene hexachloride.

The product is also called gammaxene or BHC, C6H6Cl6.

It was once widely used as an insecticide.

Q5(d). What happens when benzene is burnt in a free supply of air?

Benzene burns with a smoky flame because of its high carbon content.

In a free supply of air it is completely oxidised to carbon dioxide and water.

The equation is: 2C6H6 + 15O2 gives 12CO2 + 6H2O.

In a limited supply of air, it gives carbon monoxide or soot instead.

## Exercise Q1 — Fill in the blanks (continued: ix–x)

The molecular formula of C6H6 indicates that it is a highly unsaturated compound.

On oxidation in the presence of V2O5 benzene gives maleic acid (the ring is destroyed on strong heating with air over V2O5).

**Exercise Q2 — Indicate True or False**

Benzene is more reactive than alkene and less reactive than alkane.

False

Benzene has a pentagonal structure.

False

The C-C bond length in benzene molecule is 1.397 A.

True

The state of hybridization of carbon in benzene molecule is sp3.

False

There are six sigma bonds in benzene molecule.

False

Halogenonium ion produced in electrophilic substitution reactions is a powerful electrophile.

True

In electrophilic substitution reactions, addition products are favourable.

False

Sulphonation is carried out when benzene is heated with conc. HNO3.

False

In ozonolysis benzene directly gives glyoxal.

True

Benzene has five resonance contributing structures.

True

Exercise Q3 — Multiple choice questions (continued)

The benzene molecule contains:

A. Three double bonds

B. Two double bonds

C. One double bond

✓ D. Delocalized pi-electron charge

Benzene cannot undergo:

A. substitution reactions

B. addition reactions

C. oxidation reactions

✓ D. elimination reactions

Amongst the following, the compound that can be most readily sulphonated is:

✓ A. toluene

B. benzene

C. nitrobenzene

D. chlorobenzene

During nitration of benzene, the active nitrating agent is:

A. NO3

✓ B. NO2+

C. NO2-

D. HNO3

Which compound is the most reactive one:

A. benzene

✓ B. ethene

C. ethane

D. ethyne

The electrophile in aromatic sulphonation is:

A. $H_2SO_4$

B. $HSO_4^-$

✓ C. $SO_3$

D. $SO_3^+$

Aromatic compounds burn with sooty flame because:

A. They have high percentage of hydrogen.

B. They have a ring structure.

✓ C. They have high percentage of carbon.

D. They resist reaction with air.

The conversion of n-hexane into benzene by heating in the presence of Pt is called:

A. Isomerization

✓ B. Aromatization

C. Dealkylation

D. Rearrangement

**Exercise Q5(c), Q6 — Short questions (continued)**

Q5(c). What happens when a mixture of benzene vapours and air is passed over heated vanadium pentoxide?

Benzene is oxidised by air over heated $V_2O_5$ and the ring is destroyed.

The product is maleic acid (maleic anhydride under the reaction conditions).

This is catalytic oxidation, unlike the resistance of benzene to $KMnO_4$ or $K_2Cr_2O_7$ at room temperature.

Q6(i). What is meant by the term aromatic?

Aromatic originally meant having a pleasant smell, from the Greek word aroma.

In modern chemistry it means a compound containing a benzene ring with delocalised pi electrons.

Such compounds are unusually stable and undergo electrophilic substitution rather

than addition.

Q6(ii). What is meant by oxidation?

Oxidation is a reaction in which a substance loses hydrogen or gains oxygen.

For benzene, ordinary oxidising agents do not attack the ring at room temperature.

Strong heating with air over V2O5 destroys the ring and gives maleic acid.

Q6(iii). What is meant by sulphonation?

Sulphonation is the replacement of a ring hydrogen of benzene by a sulphonic acid group, -SO3H.

It is carried out by heating benzene with concentrated or fuming sulphuric acid.

The electrophile is sulphur trioxide, SO3, and the product is benzene sulphonic acid.

Q6(iv). What is meant by nitration?

Nitration is the replacement of a ring hydrogen by a nitro group, -NO2.

Benzene is nitrated with a mixture of concentrated nitric and sulphuric acids.

The active nitrating agent is the nitronium ion, NO2+.

Q6(v). What is meant by halogenation?

Halogenation is the introduction of a halogen atom into the benzene ring.

It is an electrophilic substitution, carried out with Cl2 or Br2 in the presence of a Lewis acid catalyst.

The product keeps the aromatic ring, with one hydrogen replaced by the halogen.

## Exercise Q7 — Structural formulas and isomers (in words)

Q7(a). Draw structural formulas for: (i) m-chlorobenzoic acid, (ii) p-hydroxybenzoic acid, (iii) o-bromonitrobenzene, (iv) o-ethyltoluene, (v) p-nitroaniline, (vi) 2,4,6-trinitrotoluene, (vii) m-nitrophenol, (viii) p-dibenzylbenzene, (ix) 2-amino-5-bromo-3-nitrobenzenesulphonic acid.

In each case the benzene ring carries the named groups at the stated positions.

(i) -COOH at position 1 and -Cl at position 3. (ii) -COOH at 1 and -OH at 4.

(iii) -Br at 1 and -NO2 at 2. (iv) -CH3 at 1 and -C2H5 at 2. (v) -NH2 at 1 and -NO2 at 4.

(vi) -CH3 at 1 with -NO2 at 2, 4 and 6. (vii) -OH at 1 and -NO2 at 3.

(viii) -CH2C6H5 groups at 1 and 4. (ix) -SO3H at 1, -NH2 at 2, -NO2 at 3 and -Br at 5.

Q7(b). Give names and the possible isomeric structures of: (i) xylenes, (ii) trimethylbenzene, (iii) bromonitrotoluene.

Xylenes are dimethylbenzenes: o-xylene (1,2-), m-xylene (1,3-) and p-xylene (1,4-dimethylbenzene).

Trimethylbenzenes are 1,2,3-, 1,2,4- and 1,3,5-trimethylbenzene (hemimellitene, pseudocumene and mesitylene).

Bromonitrotoluenes have -CH3, -Br and -NO2 on one ring in the ten possible relative positions, named by numbering the ring so the substituents get the lowest locants.

**Exercise Q9–Q12, Q14–Q15 — Short and extensive questions**

Q9. Give the general mechanism of electrophilic aromatic substitution.

First the electrophile attacks the benzene ring and forms a resonance-stabilised carbocation intermediate, the arenium ion.

The ring loses its aromatic stability in this slow, rate-determining step.

Then a base removes a proton from the intermediate, the pi system is restored, and the substituted benzene is formed.

Q10(a). Describe the structure of benzene on the basis of (i) atomic orbital treatment and (ii) resonance method.

Atomic orbital treatment: each carbon is sp2 hybridised, forming three sigma bonds in a planar hexagon.

The remaining unhybridised 2pz orbital on each carbon overlaps sideways to give a delocalised pi cloud above and below the ring.

Resonance method: benzene is a hybrid of two Kekule structures and three Dewar

structures, with a resonance energy of 150.5 kJ/mol, so no single structure describes it.

Q10(b). Prove that benzene has a cyclic structure.

Benzene, C6H6, forms only one monosubstituted product, which fits a symmetrical ring and not an open chain.

It gives three disubstituted isomers (ortho, meta, para), exactly as a six-membered ring predicts.

Ozonolysis gives three molecules of glyoxal, showing three double bonds arranged in a ring.

Q11. Predict the major products of bromination of: (a) toluene, (b) nitrobenzene, (c) bromobenzene, (d) benzoic acid, (e) benzaldehyde, (f) phenol.

Ortho- and para-directing groups give mainly o- and p-bromo products: (a) o- and p-bromotoluene, (c) o- and p-dibromobenzene, (f) o- and p-bromophenol.

Meta-directing groups give mainly the m-bromo product: (b) m-bromonitrobenzene, (d) m-bromobenzoic acid, (e) m-bromobenzaldehyde.

The directing effect comes from whether the group donates electron density to the ring or withdraws it.

Q12. How will you prepare from benzene in two steps: (a) m-chloronitrobenzene, (b) p-chloronitrobenzene?

For (a), first nitrate benzene to nitrobenzene, then chlorinate: the nitro group is meta-directing, so chlorine enters the meta position.

For (b), first chlorinate benzene to chlorobenzene, then nitrate: the chlorine atom is ortho- and para-directing, and the para product is separated from the ortho product.

The order of the two steps controls the position, because the first group directs the second.

Q14. Detail three reactions in which benzene behaves as if it is a saturated hydrocarbon and three in which it behaves as if it is unsaturated.

As if saturated, benzene undergoes substitution with the ring intact: nitration to nitrobenzene, sulphonation to benzene sulphonic acid, and halogenation to chlorobenzene or bromobenzene.

As if unsaturated, benzene undergoes addition: catalytic hydrogenation to cyclohexane, addition of chlorine in sunlight to benzene hexachloride, and ozonolysis to glyoxal.

Addition needs drastic conditions, which shows the extra stability of the aromatic ring.

Q15. What are Friedel-Crafts reactions? Give the mechanism with examples of (i) alkylation and (ii) acylation.

Friedel-Crafts reactions are electrophilic substitutions of benzene catalysed by anhydrous AlCl3.

In alkylation, an alkyl halide with AlCl3 gives a carbocation electrophile, which substitutes a ring hydrogen; for example, benzene with CH3Cl and AlCl3 gives toluene.

In acylation, an acid chloride with AlCl3 gives an acylium ion electrophile; for example, benzene with CH3COCl and AlCl3 gives acetophenone.

In both cases the arenium ion intermediate loses a proton to restore the aromatic ring.

## Exercise Q8 and Q13

Q8. Write IUPAC names of the molecules (book p.37). Q13. Complete the reactions, with conditions (book p.38).

Q8: (a) 4-bromo-2,6-dimethylbenzaldehyde — the aldehyde carbon is carbon 1, methyl groups at carbons 2 and 6, bromine at carbon 4.

(b) 3-bromo-4-ethyltoluene — the methyl group is carbon 1, bromine at carbon 3, ethyl at carbon 4.

(c) 2-bromo-4-chloro-6-iodophenol — the hydroxyl group is carbon 1, bromine at carbon 2, chlorine at carbon 4, iodine at carbon 6.

Q13: (1) Benzene + hydrogen over a nickel catalyst gives cyclohexane.

(2) Benzene + oxygen, burning in a free supply of air, gives carbon dioxide and water.

(3) Phenol heated with zinc dust gives benzene (and zinc oxide).

(4) Benzenesulphonic acid + steam (superheated steam) gives benzene and sulphuric acid — desulphonation.

(5) Benzene + sulphur trioxide in concentrated sulphuric acid (fuming sulphuric acid) gives benzenesulphonic acid.

## Chapter 10: Alkyl Halides

### Exercise Q1 — Fill in the blanks (selection)

Monohalo derivatives of alkanes are called alkyl halides .

The general formula of alkyl halides is $C_nH_{2n+1}X$ .

An alkyl group with a partial positive charge on the carbon atom is called a carbocation centre.

Alkyl halides undergo nucleophilic substitution reactions with nucleophiles.

### Exercise — Short questions (selection)

What are alkyl halides? How are they classified?

Alkyl halides are compounds formed when a hydrogen atom of an alkane is replaced by a halogen.

They are classified as primary, secondary or tertiary, according to the carbon holding the halogen.

Primary halides have the halogen on a carbon attached to one other carbon.

Secondary and tertiary halides have it on carbons attached to two and three carbons respectively.

Why are alkyl halides more reactive than alkanes?

The carbon-halogen bond is polar because the halogen is more electronegative than carbon.

The carbon therefore carries a partial positive charge and is attacked by nucleophiles.

The halide ion is also a good leaving group, so substitution takes place easily.

Alkanes have no polar bond and no leaving group, so they are much less reactive.

Distinguish between SN1 and SN2 reactions.

SN1 is a two-step reaction through a carbocation intermediate, and its rate depends only on the alkyl halide.

Tertiary halides favour SN1 because their carbocations are stable.

SN2 is a one-step reaction in which the nucleophile attacks from the back side as the halide leaves.

Primary halides favour SN2, and the product has inverted configuration.

**Exercise Q1 — Fill in the blanks (book wording)**

In tertiary alkyl halides the halogen atom is attached to a carbon which is further attached to three carbon atoms directly.

The best method for the preparation of alkyl halides is the reaction of alcohols with inorganic reagents.

An alkyl group with a partial positive charge on the carbon atom is called an electrophilic centre.

The mechanism is called SN1 if it involves one molecule in the rate-determining step.

Molecularity of a reaction is defined as the number of molecules taking part in the reaction .

Molecularity of E2 reactions is always two and the reactions show second order kinetics.

Wurtz synthesis is useful for the preparation of higher (symmetrical) alkanes.

Grignard reagents are prepared by the reaction of magnesium metal with alkyl halides in the presence of dry ether .

**Exercise Q2 — Indicate True or False**

In secondary alkyl halides, the halogen atom is attached to a carbon which is further attached to two carbon atoms directly.

True

Alcohols react with thionyl chloride in ether as solvent to give alkyl halides.

True

Order of reactivity of alkyl halides for a particular alkyl group is: Iodide > Bromide > Chloride > Fluoride.

True

In SN2 reactions the attacking nucleophile always attacks from the side on which the leaving group is attached.

False

Methyl magnesium iodide on hydrolysis yields ethyl alcohol.

False

Primary, secondary and tertiary amines react with Grignard reagents in the same way.

False

The reactions of secondary alkyl halides may follow both SN1 and SN2 mechanisms.

True

SN1 mechanism is a one stage process involving a simultaneous bond breakage and bond formation.

False

In beta-elimination reactions, the two atoms or groups attached to two adjacent carbon atoms are lost under the influence of an electrophile.

False

The reactivity order of alkyl halides is determined by the strength of carbon-halogen bond.

True

## Exercise Q3 — Multiple choice questions

In primary alkyl halides, the halogen atom is attached to a carbon which is further attached to how many carbon atoms?

A. Two

B. Three

✓ C. One

D. Four

The reactivity order of alkyl halides for a particular alkyl group is:

A. Fluoride > Chloride > Bromide > Iodide

B. Chloride > Bromide > Fluoride > Iodide

✓ C. Iodide > Bromide > Chloride > Fluoride

D. Bromide > Iodide > Chloride > Fluoride

When $CO_2$ is made to react with ethyl magnesium iodide, followed by acid hydrolysis, the product formed is:

A. Propane

✓ B. Propanoic acid

C. Propanal

D. Propanol

Grignard reagent is reactive due to:

A. the presence of halogen atom

B. the presence of Mg atom

✓ C. the polarity of C-Mg bond

D. none of the above

SN2 reactions can be best carried out with:

✓ A. Primary alkyl halides

B. Secondary alkyl halides

C. Tertiary alkyl halides

D. All the three

Elimination bimolecular reactions involve:

A. first order kinetics

✓ B. second order kinetics

C. third order kinetics

D. zero order kinetics

For which mechanisms, the first step involved is the same?

A. E1 and E2

B. E2 and SN2

C. SN1 and E2

✓ D. E1 and SN1

Alkyl halides are considered to be very reactive compounds towards nucleophiles, because:

A. they have an electrophilic carbon

✓ B. they have an electrophilic carbon and a good leaving group

C. they have an electrophilic carbon and a bad leaving group

D. they have a nucleophilic carbon and a good leaving group

The rate of E1 reaction depends upon:

✓ A. the concentration of substrate

B. the concentration of nucleophile

C. the concentration of substrate as well as nucleophile

D. None of the above

Which one of the following is not a nucleophile:

A. H2O

B. H2S

✓ C. BF3

D. NH3

## Exercise Q1–Q5, Q7–Q12

Q4. Define alkyl halide. Which is the best method of preparing alkyl halides?

An alkyl halide is a compound formed when a hydrogen atom of an alkane is replaced by a halogen atom.

The best method of preparation is the reaction of alcohols with inorganic halides such as SOCl2, PX3 or PX5.

These reagents replace -OH by -X cleanly, and the by-products escape as gases or are easily removed.

Q5. Write down a method for the preparation of ethyl magnesium bromide in the laboratory?

Dry ethyl bromide is added to magnesium turnings in dry ether, with stirring and gentle warming to start the reaction.

The magnesium inserts into the C-Br bond to give ethyl magnesium bromide, C2H5MgBr, dissolved in ether.

Water must be strictly excluded, because water destroys the Grignard reagent by hydrolysis.

Q6. Give IUPAC names to the compounds.

The text-given compounds are named: (iv) 1-chloro-2,2-dimethylpropane, (v) 2-bromopropane, (vi) 1-chloro-2-ethyl-4-methylpentane, (vii) tetrabromomethane, (ix) dichloromethane, (x) 1-bromo-4,4-dimethylpentane, (xiii) 3-bromo-3-ethylpentane.

In each name the longest chain containing the halogen is numbered to give the halogen the lowest locant.

Diagram items, read from book p.23: (i) CH3-CH(CH3)-CH2-CHCl-CH3 is 2-chloro-4-methylpentane. (ii) CH3CH2-CH(CH3)-CHBr-CH2CH3 is 3-bromo-4-methylhexane. (iii) (CH3CH2)2CH-CH2-CHCl-CH3 is 2-chloro-4-ethylhexane. (viii) BrCH2-CH2Br is 1,2-dibromoethane. (xi) (CH3)2CH-CH2-CCl(CH3)2 is 2-chloro-2,4-dimethylpentane. (xii) (CH3)2CCl-CCl(CH3)2 is 2,3-dichloro-2,3-dimethylbutane.

Q7. Draw all the possible structures that have the molecular formula C6H13Cl. Classify each as primary, secondary or tertiary chloride and give IUPAC names.

There are 17 structural isomers of C6H13Cl. Primary chlorides: 1-chlorohexane, 1-chloro-2-methylpentane, 1-chloro-4-methylpentane, 1-chloro-3-methylpentane, 1-chloro-2-ethylbutane, 1-chloro-2,2-dimethylbutane, 1-chloro-3,3-dimethylbutane, 1-chloro-2,3-dimethylbutane.

Secondary chlorides: 2-chlorohexane, 3-chlorohexane, 3-chloro-2-methylpentane, 2-chloro-4-methylpentane, 2-chloro-3-methylpentane, 3-chloro-2,2-dimethylbutane.

Tertiary chlorides: 2-chloro-2-methylpentane, 3-chloro-3-methylpentane, 2-chloro-2,3-dimethylbutane.

Classification follows the carbon carrying chlorine: attached to one, two or three other carbons respectively.

Q8. Using ethyl bromide as starting material, prepare: (a) n-butane, (b) ethyl alcohol, (c) ethyl cyanide, (d) ethane, (e) ethene, (f) propanoic acid, (g) propane.

(a) Wurtz reaction: 2 C2H5Br + 2 Na gives n-butane. (b) Aqueous KOH gives ethyl alcohol. (c) KCN gives ethyl cyanide (propanenitrile).

(d) Reduction with Zn and acid gives ethane. (e) Alcoholic KOH gives ethene by elimination. (f) KCN followed by hydrolysis of the nitrile gives propanoic acid.

(g) Through the Grignard reagent with methyl bromide, C2H5MgBr + CH3Br gives propane.

Q9. Write a detailed note on the mechanism of nucleophilic substitution reactions.

Nucleophilic substitution replaces the halogen of an alkyl halide by a nucleophile.

In SN2, the nucleophile attacks from the back side in one step, through a transition state, with inversion of configuration; the rate depends on both reactants.

In SN1, the C-X bond first breaks to give a planar carbocation, which is then attacked from either side; the rate depends only on the alkyl halide, and racemisation occurs.

Primary halides favour SN2 and tertiary halides favour SN1, while secondary halides may follow either path.

Q10. What do you understand by beta-elimination? Explain briefly the two possible mechanisms.

Beta-elimination removes a hydrogen from the beta carbon and the halogen from the alpha carbon, forming a double bond.

In E2, the base removes the beta hydrogen and the leaving group leaves in one concerted step; the rate depends on both substrate and base.

In E1, the leaving group first leaves to give a carbocation, and then the base removes a beta hydrogen; the rate depends only on the substrate.

E1 and SN1 share the same first step, so they often compete with each other.

Q11. What products are formed when the following are treated with ethyl magnesium bromide followed by acid hydrolysis: (i) HCHO, (ii) CH3CHO, (iii) CO2, (iv) (CH3)2CO, (v) CH3CH2CHO, (vi) ClCN?

Ethyl magnesium bromide adds to the carbonyl carbon, and acid hydrolysis gives the alcohol.

(i) HCHO gives propan-1-ol. (ii) CH3CHO gives butan-2-ol. (iii) CO2 gives propanoic acid.

(iv) (CH3)2CO gives 2-methylbutan-2-ol. (v) CH3CH2CHO gives pentan-3-ol. (vi) ClCN gives propanenitrile.

Formaldehyde gives a primary alcohol, other aldehydes a secondary alcohol, and ketones a tertiary alcohol.

Q12. How will you carry out the following conversions: (i) CH4 to CH3CH2COOH, (ii) CH3CH3 to (CH3CH2)4N+ Br-, (iii) CH2=CH2 to CH3CH2CH2CH2OH, (iv) CH3CH2CH2Cl to CH3CH=CH2, (v) CH3COOH to CH3CH2COOH?

(i) Methane to propanoic acid: CH4 to CH3Cl, then to CH3MgCl, then with formaldehyde to ethanol, then to bromoethane, then to C2H5MgBr, then with CO2 and hydrolysis to propanoic acid.

(ii) Ethane to tetraethylammonium bromide: ethane to bromoethane, then stepwise with ammonia to triethylamine, then with bromoethane to (C2H5)4N+ Br-.

(iii) Ethene to butan-1-ol: ethene to bromoethane, then to C2H5MgBr, then with ethylene oxide and hydrolysis to butan-1-ol.

(iv) 1-Chloropropane to propene: alcoholic KOH causes dehydrohalogenation to propene.

(v) Acetic acid to propanoic acid: acetic acid reduced to ethanol, then to bromoethane, then to C2H5MgBr, then with CO2 and hydrolysis to propanoic acid.

## Chapter 11: Alcohols, Phenols and Ethers

### Exercise Q2 — Indicate True or False (selection)

Methylated spirit contains 95% methyl alcohol and 5% ethyl alcohol.

False

Phenol is more acidic than alcohol.

True

Ethers are less reactive than alcohols.

True

Ethanol is used as a fuel in some countries.

True

**Exercise — Short questions (selection)**

Why is phenol more acidic than alcohol?

When phenol loses a proton it forms the phenoxide ion, which is stabilised by resonance.

The negative charge is spread over the benzene ring, so the ion is stable.

When an alcohol loses a proton, the alkoxide ion has no such resonance stabilisation.

Phenol therefore ionises more readily and is a stronger acid than alcohol.

How are alcohols classified? Give one example of each class.

Alcohols are classified by the number of hydroxyl groups: monohydric, dihydric and trihydric.

Ethanol is monohydric, ethylene glycol is dihydric and glycerol is trihydric.

They are also classified as primary, secondary or tertiary by the carbon holding the OH group.

Methanol is primary, isopropyl alcohol is secondary and tertiary butyl alcohol is tertiary.

Why do alcohols have higher boiling points than the corresponding alkanes?

Alcohol molecules are linked by hydrogen bonds between the oxygen of one

molecule and the hydrogen of another.

These bonds need extra energy to break, so more heat is required to boil the liquid.

Alkanes have only weak van der Waals forces between their molecules.

Alcohols therefore boil at much higher temperatures than alkanes of similar molar mass.

**Exercise Q1 — Fill in the blanks**

Primary, secondary and tertiary alcohols can be identified by Lucas test.

Oxidation of secondary alcohols give ketones.

Alcohols on heating with conc. $H_2SO_4$ give alkenes at high temperature.

Alcohols have higher boiling points than ethers due to stronger hydrogen bonding.

Williamsons synthesis is used to prepare ethers .

Methanol is also called wood spirit.

Carbolic acid is the other name of phenol.

Primary, secondary and tertiary alcohols can be prepared by reacting Grignard reagent with formaldehyde, other aldehydes and ketones .

Alcohols and carboxylic acids react to produce esters.

Ethylene glycol is used as anti-freezing agent in automobile radiator.

The process of conversion of starch into alcohol with the help of microorganisms fermentation is called.

Ketones on reduction give secondary alcohols.

**Exercise Q2 — Indicate True or False (continued)**

Ethyl alcohol is a very good anti-freezing agent.

True

Methanol is also called wood spirit.

True

Only 14% ethyl alcohol can be prepared by fermentation.

False

Ethers do not show hydrogen bonding.

True

Phenol is more soluble in water than lower alcohols.

False

Alcohols are more basic than ethers.

False

Ethers have higher boiling points than alcohols and phenols.

False

Methanol and ethanol can be distinguished by iodoform test.

True

**Exercise Q3 — Multiple choice questions (Q3 viii omitted as uncertain — see flagged block)**

Which compound shows hydrogen bonding?

A. C2H6

B. C2H5Cl

C. CH3-O-CH3

✓ D. C2H5OH

Which compound shows maximum hydrogen bonding with water?

✓ A. CH3OH

B. C2H5OH

C. CH3-O-CH3

D. C6H5OH

Which compound is more soluble in water?

✓ A. C2H5OH

B. C6H5OH

C. CH3COCH3

D. n-Hexanol

Which compound will have the maximum repulsion with H2O?

✓ A. C6H6

B. C2H5OH

C. CH3CH2CH2OH

D. CH3-O-CH3

Ethanol can be converted into ethanoic acid by:

A. Hydrogenation

B. Hydration

✓ C. Oxidation

D. Fermentation

Which enzyme is not involved in fermentation of starch?

A. Diastase

B. Zymase

✓ C. Urease

D. Invertase

Which compound is called a universal solvent?

✓ A. H2O

B. CH3OH

C. C2H5OH

D. CH3-O-CH3

Rectified spirit contains alcohol about:

A. 80%

B. 85%

C. 90%

✓ D. 95%

According to Lewis concept ethers behave as:

A. Acid

✓ B. Base

C. Acid as well as a base

D. None of them

**Exercise Q1–Q19**

Q4. What are alcohols? How are they classified? How will you distinguish between primary, secondary and tertiary alcohols?

Alcohols are hydroxy derivatives of aliphatic hydrocarbons, with general formula R-OH.

They are classified as monohydric (one -OH), dihydric (two -OH) and trihydric (three -OH); also as primary, secondary or tertiary by the carbon carrying -OH.

Lucas reagent (conc. HCl with ZnCl2) distinguishes them: tertiary alcohols give turbidity at once, secondary after a few minutes, and primary only on heating.

Q5. How is methyl alcohol obtained on a large scale? How may it be distinguished from ethyl alcohol?

Methanol is made on a large scale from water gas: CO + 2 H2 gives CH3OH over a ZnO/Cr2O3 catalyst at high temperature and pressure.

Methanol and ethanol are distinguished by the iodoform test: ethanol gives a yellow precipitate of iodoform, methanol does not.

Ethanol on oxidation gives acetaldehyde and then acetic acid, while methanol gives formaldehyde and then formic acid.

Q6. What is fermentation? Which compound may be obtained on industrial scale by fermentation?

Fermentation is the breakdown of carbohydrates by microorganisms (yeast) with

enzymes, giving alcohol and carbon dioxide.

Starch is first converted by diastase to maltose, then by maltase to glucose, and zymase converts glucose to ethyl alcohol and carbon dioxide.

Ethyl alcohol is the compound obtained on an industrial scale by fermentation, for example in Pakistan from molasses.

Q7. Explain: absolute alcohol, methylated spirit, rectified spirit, denaturing of alcohols.

Absolute alcohol is almost 100% pure ethyl alcohol, free from water.

Rectified spirit is about 95% ethyl alcohol and 5% water, the mixture obtained by ordinary distillation.

Methylated spirit is rectified spirit denatured by adding methanol (and a dye) so that it is unfit for drinking.

Denaturing means adding a poisonous or bad-tasting substance to alcohol to make it undrinkable.

Q8. How does ethyl alcohol react with: (i) conc. $H_2SO_4$, (ii) Na, (iii) $PCl_5$, (iv) $CH_3COOH$, (v) $SOCl_2$?

(i) With concentrated $H_2SO_4$ at 170 C, ethyl alcohol dehydrates to ethene; at 140 C it gives diethyl ether.

(ii) With sodium it gives sodium ethoxide and hydrogen gas. (iii) With $PCl_5$ it gives chloroethane, $POCl_3$ and HCl.

(iv) With acetic acid it gives ethyl acetate (esterification). (v) With $SOCl_2$ it gives chloroethane, $SO_2$ and HCl.

Q9. How will you obtain primary, secondary and tertiary alcohols by reacting Grignard reagent with suitable carbonyl compounds?

A Grignard reagent adds to the carbonyl carbon, and acid hydrolysis gives an alcohol.

With formaldehyde it gives a primary alcohol (for example, $CH_3MgI$ with HCHO gives

ethanol).

With any other aldehyde it gives a secondary alcohol, and with a ketone it gives a tertiary alcohol.

Q10. How will you distinguish between: (i) an alcohol and a phenol, (ii) an alcohol and an ether, (iii) methanol and ethanol, (iv) a tertiary alcohol and a primary alcohol, (v) 1-propanol and 2-propanol?

(i) An alcohol gives a neutral ferric chloride test result and no bromine-water precipitate, while phenol gives a violet colour with FeCl3 and a white precipitate of tribromophenol with bromine water.

(ii) An alcohol is soluble in water and reacts with sodium, while an ether does not react with sodium and is less soluble.

(iii) Methanol gives no iodoform test, while ethanol gives a yellow iodoform precipitate.

(iv) A tertiary alcohol gives immediate turbidity with Lucas reagent, while a primary alcohol reacts only on heating.

(v) 1-Propanol is a primary alcohol and reacts slowly with Lucas reagent, while 2-propanol is secondary and reacts within minutes.

Q11. Give reasons for: (i) ethyl alcohol is a liquid while methyl chloride is a gas, (ii) ethanol has higher boiling point than diethyl ether, (iii) absolute alcohol cannot be prepared by fermentation, (iv) ethanol gives different products with conc. H2SO4 under different conditions, (v) water has higher boiling point than ethanol.

(i) Ethyl alcohol is a liquid because its molecules are held together by hydrogen bonding, while methyl chloride has no hydrogen bonding and is a gas.

(ii) Ethanol has strong hydrogen bonding between its molecules, while diethyl ether molecules cannot hydrogen-bond with each other, so ethanol boils higher.

(iii) Fermentation stops when the alcohol concentration reaches about 12%, because higher concentrations kill the yeast; absolute alcohol therefore needs

distillation and chemical drying.

(iv) With concentrated $H_2SO_4$, ethanol gives ethene at 170 C and diethyl ether at 140 C, because temperature controls which dehydration path dominates.

(v) Water forms a more extensive hydrogen-bond network than ethanol, since each water molecule can form more hydrogen bonds, so water boils higher.

Q12. How will you convert: (i) methanol into ethanol, (ii) ethanol into methanol, (iii) ethanol into isopropyl alcohol, (iv) formaldehyde into ethyl alcohol, (v) acetone into ethyl alcohol?

(i) Methanol to ethanol: methanol to iodomethane, then to the Grignard reagent, then with formaldehyde and hydrolysis to ethanol.

(ii) Ethanol to methanol: ethanol oxidised to acetic acid, then decarboxylated with soda lime to methane, then chlorinated to chloromethane, then hydrolysed with aqueous KOH to methanol.

(iii) Ethanol to isopropyl alcohol: ethanol oxidised to acetaldehyde, then reacted with methylmagnesium iodide and hydrolysed to propan-2-ol.

(iv) Formaldehyde to ethyl alcohol: formaldehyde with methylmagnesium iodide, then hydrolysis, gives ethanol.

(v) Acetone to ethyl alcohol: acetone undergoes the haloform reaction to give acetate, acidified to acetic acid, then reduced with $LiAlH_4$ to ethanol.

Q13. Explain using ethyl alcohol as example: (i) oxidation, (ii) dehydration, (iii) esterification, (iv) ether formation.

(i) Oxidation: ethyl alcohol oxidised with acidified $K_2Cr_2O_7$ gives acetaldehyde, and further oxidation gives acetic acid.

(ii) Dehydration: with concentrated $H_2SO_4$ at 170 C it gives ethene, and at 140 C it gives diethyl ether.

(iii) Esterification: with acetic acid in the presence of concentrated $H_2SO_4$ it gives ethyl acetate and water.

(iv) Ether formation: two molecules of ethyl alcohol with concentrated H2SO4 at 140 C give diethyl ether and water.

Q14. Compare the reactions of phenol with those of ethanol. Discuss the difference, if any.

Phenol is more acidic than ethanol because the phenoxide ion is stabilised by resonance over the benzene ring, while the ethoxide ion is not.

Phenol reacts with NaOH to give sodium phenoxide, while ethanol does not react with aqueous NaOH.

Phenol gives a violet colour with ferric chloride and a white precipitate with bromine water, while ethanol gives neither test.

Both give esters and ethers, but phenol's reactions are those of a weak acid, whereas ethanol behaves as a very weak acid and a neutral alcohol.

Q15. Arrange in order of increasing acid strength and give reasons: H2O, C2H5OH, C6H5OH, C6H5COOH.

Increasing acid strength: C2H5OH < H2O < C6H5OH < C6H5COOH.

Ethanol is the weakest acid because the ethoxide ion is not stabilised; water is stronger than ethanol but weaker than phenol.

Phenol is stronger than water because the phenoxide ion is resonance-stabilised over the ring.

Benzoic acid is the strongest because the carboxylate ion spreads its negative charge equally over two oxygen atoms.

Q16. Write down two methods for preparing phenol. What is the action of HNO3, NaOH, Zn and bromine water on phenol?

Phenol is prepared by the Dow process (chlorobenzene with NaOH at high temperature and pressure, then acidification) and by fusion of sodium benzene sulphonate with NaOH followed by acidification.

With concentrated HNO3 it gives picric acid (2,4,6-trinitrophenol). With NaOH it

gives sodium phenoxide. With zinc dust it is reduced to benzene.

With bromine water it gives a white precipitate of 2,4,6-tribromophenol.

Q17. Give the uses of phenols. How is bakelite prepared from it?

Phenol is used as an antiseptic and disinfectant, in making bakelite, dyes, drugs and picric acid.

Bakelite is prepared by the condensation of phenol with formaldehyde in the presence of an acid or base catalyst.

The product is a hard, heat-resistant cross-linked resin used for electrical fittings and handles.

Q18. (a) Write IUPAC names of the compounds given. (b) Write structure formulas in words for: glycol, glycerol, carbolic acid, acetophenone, picric acid.

(a) (CH3)2CH-OH is propan-2-ol. (CH3)2CHCH2OH is 2-methylpropan-1-ol. (CH3)3COH is 2-methylpropan-2-ol. C2H5-CH(OH)-CH3 is butan-2-ol.

(b) In words: glycol is HO-CH2-CH2-OH; glycerol is HO-CH2-CH(OH)-CH2-OH; carbolic acid is C6H5-OH; acetophenone is C6H5-CO-CH3; picric acid is 2,4,6-trinitrophenol.

Q19. (a) Name the ethers given. (b) Write down structural formulas in words for: methoxyethane, ethoxybenzene, sodium ethoxide, sodium phenoxide, propoxypropane.

(a) CH3-CH2-CH2-O-CH3 is 1-methoxypropane. C6H5-O-C6H5 is diphenyl ether (phenoxybenzene). (CH3)2CH-O-CH(CH3)2 is diisopropyl ether (2-propoxypropane). CH3-O-C6H5 is methoxybenzene (anisole). CH3-CH2-CH2-O-CH2-CH3 is 1-ethoxypropane.

(b) In words: methoxyethane is CH3-O-CH2-CH3; ethoxybenzene is C6H5-O-C2H5; sodium ethoxide is C2H5-ONa; sodium phenoxide is C6H5-ONa; propoxypropane is CH3-CH2-CH2-O-CH2-CH2-CH3.

Methyl alcohol is not used:

A. as a solvent

B. as an anti-freezing agent

✓ C. as a substitute for petrol

D. for denaturing of ethyl alcohol

## Chapter 12: Aldehydes and Ketones

### Exercise Q1 — Fill in the blanks (selection)

Aldehydes are the first oxidation product of primary alcohols .

The carbon atom of a carbonyl group is sp2 hybridized.

Ketones are the oxidation products of secondary alcohols .

Formaldehyde is used as a disinfectant and preservative .

### Exercise Q3 — Multiple choice questions (selection)

The carbon atom of a carbonyl group is:

A. sp hybridized

✓ B. sp2 hybridized

C. sp3 hybridized

D. dsp2 hybridized

### Exercise — Short questions (selection)

Distinguish between aldehydes and ketones.

An aldehyde has the carbonyl group at the end of the chain, bonded to at least one hydrogen.

A ketone has the carbonyl group in the middle, bonded to two carbon groups.

Aldehydes are easily oxidised to carboxylic acids, while ketones resist oxidation.

Aldehydes give a silver mirror with Tollens reagent, but ketones do not.

Why is the carbonyl carbon attacked by nucleophiles?

Oxygen is more electronegative than carbon, so the C=O bond is polar.

The carbonyl carbon carries a partial positive charge.

Nucleophiles, which are electron-rich, are attracted to this positive centre.

Addition of the nucleophile to the carbon is therefore the characteristic reaction of carbonyl compounds.

**Exercise Q1 — Fill in the blanks (continued)**

Aldehydes and ketones undergo nucleophilic addition reactions.

Formaldehyde reacts with Grignard reagent to give primary alcohol.

Acetaldehyde reacts with ethyl magnesium iodide to give 2-butanol.

Aldehydes are strong reducing agents.

The oxidation of an aldehyde always gives a carboxylic acid.

The reduction of a ketone always gives a secondary alcohol.

Formaldehyde gives silver mirror test with Tollen's reagent.

Acetaldehyde gives a red precipitate with Fehling's solution.

**Exercise Q2 — Indicate True or False**

Formaldehyde is used in the silvering of mirrors.

True

Ketones combine with alcohols in the presence of HCl gas to form acetals.

False

Acetaldehyde undergoes Cannizzaro's reaction.

False

Aldol condensation reaction is given by only those aldehydes and ketones which contain an alpha-hydrogen atom.

True

Cannizzaro's reaction is given by only those aldehydes containing no alpha-hydrogen atom.

True

Propanal and propanone behave differently with Tollen's reagent.

True

Acetone reacts with sodium bisulphite to give a yellow crystalline product.

False

Acetone on reduction gives a primary alcohol.

False

40% aqueous solution of formaldehyde is called formalin.

True

**Exercise Q3 — Multiple choice questions (continued)**

Formalin is:

A. 10% solution of formaldehyde in water

B. 20% solution of formaldehyde in water

✓ C. 40% solution of formaldehyde in water

D. 60% solution of formaldehyde in water

Which of the following will have the highest boiling point?

A. Methanal

B. Ethanal

C. Propanal

✓ D. 2-Hexanone

Ketones are prepared by the oxidation of:

A. Primary alcohol

✓ B. Secondary alcohol

C. Tertiary alcohol

D. all of these

Acetone reacts with HCN to form a cyanohydrin. It is an example of:

A. Electrophilic addition

B. Electrophilic substitution

✓ C. Nucleophilic addition

D. Nucleophilic substitution

Which of the following compounds will not give iodoform test on treatment with I2/NaOH:

A. Acetaldehyde

B. Acetone

C. Butanone

✓ D. 3-Pentanone

Which of the following compounds will react with Tollen's reagent?

✓ A. CH3-CHO (ethanal)

B. CH3-CO-CH3 (propanone)

C. CH3-COOH (ethanoic acid)

D. CH3-CO-CH2-CH3 (butanone)

Cannizzaro's reaction is not given by:

A. Formaldehyde

✓ B. Acetaldehyde

C. Benzaldehyde

D. Trimethylacetaldehyde

Which of the following reagents will react with both aldehydes and ketones?

✓ A. Grignard reagent

B. Tollen's reagent

C. Fehling's reagent

D. Benedict's reagent

## Exercise Q1–Q20 (continued)

Q4. Give one laboratory and one industrial method for the preparation of formaldehyde.

Laboratory method: oxidation of methanol vapour with air over a heated platinum or copper catalyst gives formaldehyde.

Industrial method: methanol and air are passed over a silver catalyst at high temperature, giving formaldehyde on a large scale.

Formaldehyde is collected as an aqueous solution, formalin.

Q5. How does formaldehyde react with: (i) CH3MgI, (ii) HCN, (iii) NaHSO3, (iv) conc. NaOH, (v) NaBH4/H2O, (vi) Tollen's reagent, (vii) Fehling's reagent?

(i) CH3MgI, then hydrolysis: gives ethanol (a primary alcohol). (ii) HCN: gives formaldehyde cyanohydrin, hydroxymethanenitrile.

(iii) NaHSO3: gives a white crystalline addition product, used to purify aldehydes.

(iv) Concentrated NaOH: Cannizzaro reaction gives methanol and sodium formate.

(v) NaBH4/H2O: reduces formaldehyde to methanol. (vi) Tollen's reagent: gives a silver mirror. (vii) Fehling's solution: gives a red precipitate of Cu2O.

Q6. Give one laboratory and one industrial method for the preparation of acetaldehyde.

Laboratory method: oxidation of ethyl alcohol with acidified potassium dichromate gives acetaldehyde, distilled off as it forms.

Industrial method: hydration of ethyne with dilute H2SO4 and HgSO4 catalyst gives acetaldehyde.

The Wacker process (oxidation of ethene with PdCl2/CuCl and oxygen) is another industrial route to acetaldehyde.

Q7. How does acetaldehyde react with: (i) C2H5MgI, (ii) HCN, (iii) NaHSO3, (iv)

dilute NaOH, (v) I2/NaOH, (vi) NaBH4/H2O, (vii) NH2OH, (viii) K2Cr2O7/H2SO4?

(i) C2H5MgI, then hydrolysis: gives butan-2-ol. (ii) HCN: gives acetaldehyde cyanohydrin (2-hydroxypropanenitrile).

(iii) NaHSO3: gives a white crystalline bisulphite addition compound. (iv) Dilute NaOH: aldol condensation gives 3-hydroxybutanal (aldol).

(v) I2/NaOH: gives yellow iodoform precipitate. (vi) NaBH4/H2O: reduces acetaldehyde to ethanol.

(vii) NH2OH: gives acetaldoxime. (viii) K2Cr2O7/H2SO4: oxidises acetaldehyde to acetic acid.

Q8. Describe briefly the mechanism of nucleophilic addition to a carbonyl compound.

The carbonyl carbon is electrophilic because oxygen pulls electrons away from it.

The nucleophile attacks the carbonyl carbon, and the pi bond electrons move to oxygen, giving a tetrahedral alkoxide intermediate.

Protonation of the alkoxide oxygen gives the addition product, an alcohol derivative.

Q9. Explain with mechanism the addition of ethylmagnesium bromide to acetaldehyde. What is the importance of this reaction?

Ethylmagnesium bromide attacks the carbonyl carbon of acetaldehyde, giving a magnesium alkoxide intermediate.

Acid hydrolysis protonates the alkoxide to give butan-2-ol, a secondary alcohol.

The reaction is important because it forms a new carbon-carbon bond and lengthens the carbon chain by two carbons.

Q10. Explain with mechanism the addition of sodium bisulphite to acetone. What is the utility of this reaction?

The bisulphite ion attacks the carbonyl carbon of acetone, giving a white crystalline addition compound, acetone sodium bisulphite.

Acid or base treatment of the addition compound regenerates acetone.

The reaction is useful for purifying and identifying aldehydes and small ketones, since the addition compound can be filtered and then decomposed.

Q11. Describe with mechanism aldol condensation. Why does formaldehyde not give this reaction?

In the presence of dilute base, one acetaldehyde molecule forms an enolate ion, which attacks the carbonyl carbon of a second molecule.

The product after protonation is 3-hydroxybutanal, called aldol, which has both an alcohol and an aldehyde group.

Formaldehyde does not give this reaction because it has no alpha-hydrogen atom, so it cannot form an enolate ion.

Q12. What types of aldehydes give Cannizzaro's reaction? Give its mechanism.

Aldehydes with no alpha-hydrogen atom give Cannizzaro's reaction, for example formaldehyde and benzaldehyde.

With concentrated alkali, one molecule is oxidised to the carboxylic acid salt and another is reduced to the alcohol.

The mechanism involves hydride transfer from one aldehyde molecule to another, through a tetrahedral intermediate formed by hydroxide attack.

Q13. Explain the mechanism of the reaction of phenylhydrazine with acetone.

Phenylhydrazine attacks the carbonyl carbon of acetone as a nucleophile, giving a tetrahedral intermediate.

Loss of water from the intermediate gives acetone phenylhydrazone, (CH3)2C=N-NH-C6H5.

The reaction is a condensation: the carbonyl oxygen is lost as water and a C=N double bond is formed.

Q14. Using ethyne as starting material, how would you get acetaldehyde, acetone and ethyl alcohol?

Ethyne hydrated with dilute H2SO4 and HgSO4 gives acetaldehyde.

Acetaldehyde reduced with NaBH4 or H2 gives ethyl alcohol.

Acetaldehyde oxidised to acetic acid, converted to calcium acetate, and dry-distilled gives acetone; so all three products come from ethyne through acetaldehyde.

Q15. Give the mechanism of addition of HCN to acetone.

Cyanide ion attacks the carbonyl carbon of acetone, giving a tetrahedral alkoxide intermediate.

Protonation of the alkoxide gives acetone cyanohydrin, (CH3)2C(OH)CN.

The reaction is catalysed by base, which generates the cyanide nucleophile from HCN.

Q16. How would you bring about the conversions listed in the book (i)-(xii)?

(i) Acetone to t-butyl alcohol: acetone with CH3MgI, then hydrolysis, gives 2-methylpropan-2-ol.

(ii) Propanal to 1-propanol: reduction with NaBH4 gives propan-1-ol. (iii) Propanone to 2-propanol: reduction with NaBH4 gives propan-2-ol.

(iv) Methanal to ethanal: methanal with CH3MgI, then hydrolysis, gives ethanol; oxidation of ethanol gives ethanal.

(v) Ethanal to propanone: ethanal oxidised to acetic acid, converted to calcium acetate, dry-distilled to acetone.

(vi) Ethanal to 2-propanol: ethanal with CH3MgI, then hydrolysis, gives propan-2-ol.

(vii) Ethyne to ethanal: hydration of ethyne with H2SO4/HgSO4 gives ethanal. (viii) Ethene to ethanal: ethene oxidised by the Wacker process gives ethanal.

(ix) Ethanal to ethanol: reduction with NaBH4 gives ethanol. (x) Ethanol to 2-butanone: ethanol oxidised to ethanal, then Grignard with C2H5MgI gives butan-2-ol, oxidised to butanone.

(xi) Methanol to ethanal: methanol to iodomethane, then Grignard with formaldehyde gives ethanol, oxidised to ethanal.

(xii) Ethanol to ethanoic acid: oxidation with acidified K2Cr2O7 gives ethanoic acid.

Q17. How will you distinguish between the pairs listed in the book (i)-(vii)?

(i) Methanal and ethanal: ethanal gives iodoform with I2/NaOH, methanal does not.

(ii) Ethanal and propanone: ethanal gives a silver mirror with Tollen's reagent, propanone does not.

(iii) Ethanal and propanal: ethanal gives iodoform, propanal does not. (iv) Acetone and ethyl alcohol: acetone gives iodoform, ethyl alcohol also gives iodoform, but acetone does not react with sodium while ethyl alcohol liberates hydrogen.

(v) Butanone and 3-pentanone: butanone gives iodoform, 3-pentanone does not.

(vi) Acetaldehyde and benzaldehyde: acetaldehyde gives iodoform and aldol reaction, benzaldehyde gives neither but gives Cannizzaro reaction.

(vii) 2-Pentanone and 3-pentanone: 2-pentanone gives iodoform, 3-pentanone does not.

Q18. Discuss oxidation of (a) aldehydes and (b) ketones with: (i) K2Cr2O7/H2SO4, (ii) Tollen's reagent, (iii) Fehling's solution.

With K2Cr2O7/H2SO4, aldehydes are oxidised to carboxylic acids, while ketones resist oxidation except under drastic conditions where the chain breaks.

With Tollen's reagent, aldehydes give a silver mirror, while ketones give no reaction.

With Fehling's solution, aldehydes give a red precipitate of Cu2O, while ketones give no reaction.

Q19. Discuss reduction of (a) aldehydes and (b) ketones with: (i) NaBH4/H2O, (ii) H2/Pd.

With NaBH4/H2O, aldehydes are reduced to primary alcohols and ketones to secondary alcohols.

With H2/Pd, aldehydes are reduced to primary alcohols and ketones to secondary alcohols, under catalytic hydrogenation conditions.

Sodium borohydride is a selective reagent: it reduces the carbonyl group but leaves most other groups untouched.

Q20. Give three uses for each of formaldehyde and acetaldehyde.

Formaldehyde uses: as formalin for preserving biological specimens, as a disinfectant, and in making bakelite and other resins.

Acetaldehyde uses: in making acetic acid and acetic anhydride, in making resins and dyes, and as an intermediate for alcohols and acids.

Formaldehyde is also used in silvering mirrors, and acetaldehyde in the manufacture of chloral and paraldehyde.

## Chapter 13: Carboxylic Acids

### Exercise Q1 — Fill in the blanks (selection)

Carboxylic acids contain the carboxyl (-COOH) functional group.

Acetic acid is the main component of vinegar .

Formic acid occurs naturally in ants and stinging nettles .

### Exercise Q3 — Multiple choice questions (selection)

Which acid is used in the manufacture of synthetic fibre?

A. formic acid

B. oxalic acid

C. carbonic acid

✓ D. acetic acid

### Exercise — Short questions (selection)

Why are carboxylic acids stronger acids than alcohols and phenols?

A carboxylic acid loses a proton to give a carboxylate ion.

The negative charge in the carboxylate ion is shared equally by two oxygen atoms

through resonance.

This makes the ion very stable, so the acid ionises readily.

Alkoxide and phenoxide ions are less stabilised, so alcohols and phenols are weaker acids.

Why do carboxylic acids have high boiling points?

Carboxylic acid molecules form strong hydrogen bonds with each other.

Two molecules pair up as a dimer, held by two hydrogen bonds.

Breaking these bonds needs much energy, so the boiling point is high.

They therefore boil higher than alcohols and aldehydes of similar molar mass.

**Exercise Q1 — Fill in the blanks (continued)**

Formula of malonic acid is $CH_2(COOH)_2$ .

Methyl nitrile upon acidic hydrolysis produces acetic acid .

Melting points of carboxylic acids containing even number of carbon atoms are higher than the next lower and higher members containing odd number of carbon atoms.

Acetic acid on heating with phosphorus pentoxide produces acetic anhydride.

Acid chloride and acid anhydride are called derivatives of acid.

Pure acetic acid is called glacial acetic acid .

Formula of alanine is $CH_3CH(NH_2)COOH$ .

Proline is a cyclic amino acid.

A peptide having a molecular mass more than 10000 is called a protein .

**Exercise Q2 — Indicate True or False**

Acetic acid exists as a dimer in benzene.

True

First three aliphatic acids have fruity smells.

False

Carboxylic acids on reduction with LiAlH4 produce alkenes.

False

Acetic acid on dehydration produces CO and H2.

False

Sodium formate on heating with soda lime produces NaHCO3 and hydrogen.

False

Amino acids exist as Zwitter ion.

True

Histidine is an acidic amino acid.

False

A peptide having molecular mass upto 10000 is called protein.

False

Phthalic acid is a monocarboxylic acid.

False

Formula of glycine is CH2COOH.

False

**Exercise Q3 — Multiple choice questions (continued; Q3 iv omitted as uncertain — see flagged block)**

Acetic acid is manufactured by:

A. distillation

✓ B. fermentation

C. ozonolysis

D. esterification

A carboxylic acid contains:

A. a hydroxyl group

✓ B. a carboxyl group

C. a hydroxyl and carboxyl group

D. a carboxyl and an aldehydic group

Which reagent is used to reduce a carboxylic group to an alcohol?

A. H2/Ni

B. H2/Pt

C. NaBH4

✓ D. LiAlH4

The solution of which acid is used for seasoning of food?

A. formic acid

✓ B. acetic acid

C. benzoic acid

D. butanoic acid

Organic compounds X and Y react together to form organic compound Z. What type of compounds can X, Y and Z be?

A. alcohol, ester, acid

B. acid, ester, alcohol

C. ester, alcohol, acid

✓ D. alcohol, acid, ester

An aqueous solution of an organic compound reacts with sodium carbonate to produce carbon dioxide gas. Which one could it be?

A. CH2=CH-CH3

B. CH3-CHO

C. CH3COOC2H5

✓ D. CH3-CH2-COOH

Which of the following is not a fatty acid?

A. propanoic acid

B. acetic acid

✓ C. phthalic acid

D. butanoic acid

Acetamide is prepared by:

✓ A. heating ammonium acetate

B. heating methyl cyanide

C. heating ethyl acetate

D. the hydrolysis of methyl cyanide

**Exercise Q1–Q16**

Q4. Write down the structural formulae of: (i) valeric acid, (ii) propionic acid, (iii) oxalic acid, (iv) benzoic acid, (v) acetic anhydride, (vi) acetyl chloride. Q5. IUPAC names (book p.26).

In words: (i) valeric acid is CH3-CH2-CH2-CH2-COOH. (ii) Propionic acid is CH3-CH2-COOH. (iii) Oxalic acid is HOOC-COOH.

(iv) Benzoic acid is C6H5-COOH. (v) Acetic anhydride is (CH3CO)2O. (vi) Acetyl chloride is CH3-CO-Cl.

Q5, read from book p.26: (i) CH2(COOH)2 is propanedioic acid (malonic acid). (ii) Benzene-1,3-dicarboxylic acid (isophthalic acid) — carboxyl groups at carbons 1 and 3. (iii) H-COOH is methanoic acid (formic acid). (iv) CH3-COO-C2H5 is ethyl ethanoate (ethyl acetate). (v) and (vi) are named in the next item.

Q5 (v)-(vi). Write down the names by IUPAC system of the text-given compounds.

(v) NH2-CH2-COOH is glycine (aminoethanoic acid). (vi) HCOO-C3H7 is propyl methanoate (propyl formate).

Glycine is the simplest amino acid, and propyl formate is an ester of formic acid with propyl alcohol.

Q6. (a) How is acetic acid manufactured? What is glacial acetic acid? (b) How would

you convert acetic acid into: (i) methane, (ii) acetyl chloride, (iii) acetamide, (iv) acetic anhydride?

Acetic acid is manufactured by the fermentation of ethanol (vinegar process) and industrially by oxidation of acetaldehyde or carbonylation of methanol.

Glacial acetic acid is pure, anhydrous acetic acid, which freezes to ice-like crystals at about 16.6 C.

(b) (i) With soda lime and heat, acetic acid decarboxylates to methane. (ii) With $PCl_5$ or $SOCl_2$ it gives acetyl chloride.

(iii) With ammonia it gives ammonium acetate, which on heating gives acetamide.

(iv) With $P_2O_5$ (dehydration) it gives acetic anhydride.

Q7. (a) What are fatty acids? (b) What is vinegar? Describe how vinegar is prepared from ethanol?

Fatty acids are long-chain aliphatic monocarboxylic acids, usually with an even number of carbon atoms, for example palmitic acid and stearic acid.

Vinegar is a dilute (about 5%) aqueous solution of acetic acid.

It is prepared from ethanol by fermentation with Acetobacter bacteria, which oxidise ethanol to acetic acid in the presence of air.

Q8. How would you carry out: (i) acetic acid into acetamide, (ii) acetic acid into acetone?

(i) Acetic acid to acetamide: acetic acid with ammonia gives ammonium acetate, which on heating gives acetamide and water.

(ii) Acetic acid to acetone: acetic acid converted to calcium acetate, which on dry distillation gives acetone and calcium carbonate.

Both conversions use the carboxyl group: one replaces -OH by $-NH_2$, the other removes the carboxyl carbon as carbonate.

Q9. Write down the mechanisms of: (i) acetic acid and ethanol, (ii) acetic acid and ammonia, (iii) acetic acid and thionyl chloride.

(i) Acetic acid with ethanol, acid-catalysed: the alcohol oxygen attacks the carbonyl carbon, water is lost, and ethyl acetate is formed.

(ii) Acetic acid with ammonia: ammonium acetate forms first, and heating loses water to give acetamide.

(iii) Acetic acid with thionyl chloride: the -OH is replaced by -Cl through a chlorosulphite intermediate, giving acetyl chloride, SO2 and HCl.

Q10. What happens when the following are heated: (i) calcium acetate, (ii) sodium formate and soda lime, (iii) ammonium acetate?

(i) Calcium acetate on heating (dry distillation) gives acetone and calcium carbonate.

(ii) Sodium formate with soda lime on heating gives hydrogen gas and sodium carbonate.

(iii) Ammonium acetate on heating loses water and gives acetamide.

Q11. What are amino acids? Explain their different types with one example in each case.

Amino acids are compounds containing both an amino group and a carboxylic acid group, usually on the same (alpha) carbon.

By their side chains they are: neutral (glycine), acidic (aspartic acid, glutamic acid), basic (lysine, arginine, histidine), hydroxy (serine), and sulphur-containing (cysteine, methionine).

They are also classified as essential (must come from diet, for example lysine) and non-essential (made in the body, for example glycine).

Q12. Write a short note on acidic and basic characters of an amino acid.

An amino acid in solution exists as a zwitterion: the carboxyl group loses a proton and the amino group gains it, so the molecule carries both charges but is neutral overall.

In acid solution the amino acid acts as a base and becomes positively charged; in

alkaline solution it acts as an acid and becomes negatively charged.

At its isoelectric point the amino acid is neutral and has minimum solubility, which is used in separating amino acids.

Q13. What is a peptide bond? Write down the formula of a dipeptide.

A peptide bond is the amide linkage, -CO-NH-, formed between the carboxyl group of one amino acid and the amino group of another, with loss of water.

A dipeptide contains two amino acids joined by one peptide bond, for example glycylglycine: NH2-CH2-CO-NH-CH2-COOH.

Longer chains are polypeptides, and very large polypeptides (molecular mass above 10000) are proteins.

Q14. What are zwitter ions?

A zwitterion is a neutral molecule that carries both a positive and a negative charge at different atoms.

Amino acids form zwitterions because the acidic -COOH group donates its proton to the basic -NH2 group within the same molecule.

Zwitterions explain why amino acids have high melting points and are soluble in water but not in organic solvents.

Q15. What are alpha amino acids, proteins and peptides? How are they related?

Alpha amino acids are the building blocks in which -NH2 and -COOH are attached to the same carbon atom.

Two or more amino acids joined by peptide bonds form peptides; a dipeptide has two, a tripeptide three, and a polypeptide many amino acids.

Proteins are very large polypeptides, folded into specific shapes, which carry out the structural and catalytic work of the cell.

Q16. Study the facts (a)-(c) and answer: molecular mass of A, identities of A, X and B, five confirming reactions, one large-scale use.

1. Molecular mass of A = 2 x vapour density = 2 x 15 = 30.

2. A is formaldehyde, HCHO; X is methanol, CH3OH; B is formic acid, HCOOH.

3. Confirmations: A gives a silver mirror with Tollen's reagent and restores Schiff's reagent colour; X burns with a blue flame, gives no iodoform test, and with salicylic acid and concentrated H2SO4 gives the wintergreen smell of methyl salicylate. B turns blue litmus red, liberates CO2 with NaHCO3, gives hydrogen with sodium, and reduces Tollen's reagent and Fehling's solution.

4. Large-scale use: formaldehyde is used to make bakelite resins and as a preservative (formalin); methanol is used as a fuel and solvent; formic acid is used in leather tanning and dyeing.

Which following derivative can not be prepared directly from acetic acid:

✓ A. acetamide

B. acetyl chloride

C. acetic anhydride

D. ethyl acetate

## Chapter 14: Macromolecules

### Exercise Q1 — Fill in the blanks (selection)

The double-stranded structure of DNA was deduced by Watson and Crick in 1953 .

Proteins are polymers of amino acids .

Starch and cellulose are polymers of glucose .

Natural rubber is a polymer of isoprene .

### Exercise — Short questions (selection)

What are macromolecules? Give examples.

Macromolecules are very large molecules made by joining many small units called

monomers.

Natural examples are proteins, starch, cellulose and nucleic acids.

Synthetic examples are nylon, polyester and polythene.

They are essential for life and for modern materials.

Why can DNA preserve genetic information from generation to generation?

DNA has a double-stranded helix structure, discovered by Watson and Crick in 1953.

The two strands are complementary: adenine pairs with thymine and cytosine with guanine.

When a cell divides, each strand acts as a template for a new partner strand.

The genetic message is therefore copied exactly into the next generation.

**Exercise Q1 — Fill in the blanks (book wording)**

Macromolecules are built up from small units called monomers .

Nylon is a polyamide and terylene is a polyester .

Nylon is prepared by the reaction of adipic acid and hexamethylenediamine.

Based on their thermal properties, plastics are divided into two main classes.

Polyvinyl chloride is a thermoplastic plastic.

Glucose is stored as glycogen in the liver.

Glucose and fructose are water soluble carbohydrates.

Protein after digestion changes to amino acids .

Purine and pyrimidine are nitrogenous bases of nucleic acids.

Addition of a plasticizer increases the flexibility of the polymer.

**Exercise Q2 — Indicate True or False**

Nylon 6,6 and terylene are condensation polymers.

True

The disposal of plastics does not cause any pollution problem.

False

Fructose is a polysaccharide carbohydrate.

False

Human beings get no food nutrient from cellulose.

True

The most abundant and the most important steroid in the human body is vitamin D.

False

Enzymes are the compounds containing C, H and O only.

False

The degree of unsaturation of fats is measured by their iodine number.

True

Activity of an enzyme varies with temperature and pH.

True

Nucleic acids are biological catalysts.

False

The nucleic acids are responsible for protein synthesis in the human body.

True

## Exercise Q3 — Multiple choice questions

In which of these processes are small organic molecules made into macromolecules?

A. the cracking of petroleum fractions

B. the fractional distillation of crude oil

✓ C. the polymerization of ethene

D. the hydrolysis of proteins

Which of these polymers is an addition polymer?

A. nylon-6,6

✓ B. polystyrene

C. terylene

D. epoxy resin

Which of these polymers is a synthetic polymer?

A. animal fat

B. starch

C. cellulose

✓ D. polyester

Plastics are a pollution problem because many plastics:

A. are made from petroleum

B. are very inflammable

✓ C. burn to produce toxic fumes

D. decompose to produce toxic products

The fibre which is made from acrylonitrile as monomer:

A. PVC

B. rayon fibre

✓ C. acrylic fibre

D. polyester fibre

A polymeric substance that is formed in the liquid state and then hardened to a rigid solid is called a:

A. fibre

B. plastic

✓ C. varnish

D. polyamide resin

Vegetable oils are:

A. unsaturated fatty acids

✓ B. glycerides of unsaturated fatty acids

C. glycerides of saturated fatty acids

D. essential oils obtained from plants

Which one of the following elements is not present in all proteins?

A. carbon

B. hydrogen

C. nitrogen

✓ D. sulphur

Which one of the following nitrogenous bases is not present in RNA?

A. cytosine

B. adenine

✓ C. thymine

D. uracil

Which one of the following enzymes brings about the hydrolysis of fats?

A. urease

B. maltase

C. zymase

✓ D. lipase

The reaction between fat and NaOH is called:

A. esterification

B. hydrogenolysis

C. fermentation

✓ D. saponification

Which one of the following statements about glucose and sucrose is incorrect?

A. both are soluble in water

B. both are naturally occurring

C. both are carbohydrates

✓ D. both are disaccharides

## Exercise Q1–Q13 (continued)

Q4. Explain: (a) addition polymer, (b) condensation polymer, (c) thermoplastic, (d) thermosetting plastic.

An addition polymer is formed by repeated addition of monomer molecules across a double bond, with no by-product; for example, polyethene from ethene.

A condensation polymer is formed by monomers joining with loss of a small molecule such as water; for example, nylon 6,6 from adipic acid and hexamethylenediamine.

A thermoplastic softens on heating and hardens on cooling, and can be remoulded; a thermosetting plastic becomes permanently hard on heating and cannot be remoulded.

Q5. Write notes on: (a) polyester resins, (b) polyamide resins, (c) epoxy resins.

Polyester resins are condensation polymers of dicarboxylic acids and diols, for example terylene; they are strong, resistant fibres and fabrics.

Polyamide resins are condensation polymers of diamines and dicarboxylic acids, for example nylon; they are tough, elastic fibres used in ropes, fabrics and brushes.

Epoxy resins are cross-linked polymers formed from epoxides; they are strong adhesives and coatings, resistant to chemicals and heat.

Q6. What is the repeating unit in each of: (a) polystyrene, (b) nylon 6,6, (c) teflon, (d) orlon?

(a) Polystyrene repeating unit: -CH2-CH(C6H5)-, from styrene. (b) Nylon 6,6 repeating unit: -NH-(CH2)6-NH-CO-(CH2)4-CO-, from adipic acid and hexamethylenediamine.

(c) Teflon repeating unit: -CF2-CF2-, from tetrafluoroethene. (d) Orlon repeating unit: -CH2-CH(CN)-, from acrylonitrile.

The repeating unit is the smallest group of atoms that repeats along the polymer chain.

Q7. What are carbohydrates and how are they classified?

Carbohydrates are polyhydroxy aldehydes or ketones, or compounds that give them on hydrolysis; they are the most abundant biomolecules on earth.

They are classified as monosaccharides (glucose, fructose), oligosaccharides (sucrose, maltose, lactose) and polysaccharides (starch, cellulose, glycogen).

Monosaccharides cannot be hydrolysed further, while polysaccharides give many monosaccharide units on hydrolysis.

Q8. Point out one difference between: (a) glucose and fructose, (b) sucrose and maltose, (c) cellulose and starch.

(a) Glucose is an aldohexose (aldehyde group), while fructose is a ketohexose (keto group).

(b) Sucrose is a non-reducing sugar (glucose + fructose), while maltose is a reducing sugar (glucose + glucose).

(c) Cellulose is a straight-chain polymer of beta-glucose used for structure, while starch is a polymer of alpha-glucose used for storage, with amylose and amylopectin.

Q9. What are lipids? In what way are fats and oils different?

Lipids are naturally occurring organic compounds of plant and animal origin, soluble in organic solvents and insoluble in water.

Fats and oils are both glycerides (esters of glycerol with fatty acids); fats are solid at room temperature because they contain mostly saturated fatty acids.

Oils are liquid at room temperature because they contain mostly unsaturated fatty acids, with lower melting points.

Q10. Define saponification number and iodine number. Discuss rancidity.

Saponification number is the number of milligrams of KOH needed to saponify one gram of fat or oil; it measures the average molecular mass of the fatty acids.

Iodine number is the number of grams of iodine that add to 100 grams of fat or oil;

it measures the degree of unsaturation.

Rancidity is the unpleasant smell and taste that fats develop when they are oxidised by air or hydrolysed by moisture and enzymes, giving free fatty acids and aldehydes.

Q11. What is the difference between a glycosidic linkage and a peptide linkage?

A glycosidic linkage joins two monosaccharide units through an oxygen atom, as in sucrose, maltose, starch and cellulose.

A peptide linkage joins two amino acids through an amide bond, -CO-NH-, as in proteins.

The glycosidic linkage is an ether-type (acetal) linkage between sugars, while the peptide linkage is an amide linkage between amino acids.

Q12. What is the chemical nature of enzymes? Discuss the classification of enzymes.

Enzymes are proteins that act as biological catalysts; they are specific, work under mild conditions, and are not used up in the reaction.

They are classified by the reaction they catalyse: oxidoreductases (oxidation-reduction), transferases (group transfer), hydrolases (hydrolysis), lyases (addition or removal), isomerases (isomerisation) and ligases (bond formation with ATP).

Enzyme activity depends on temperature and pH, with an optimum for each enzyme, and enzymes are inhibited or denatured outside it.

Q13. What are nucleic acids? Write down the role of DNA and RNA in life.

Nucleic acids are polynucleotides made of nucleotide units (sugar + phosphate + nitrogenous base); DNA and RNA are the two types.

DNA is the double-stranded store of genetic information; it replicates and passes information from generation to generation.

RNA is usually single-stranded and carries out protein synthesis: messenger RNA

carries the code, transfer RNA brings amino acids, and ribosomal RNA builds proteins.

## Chapter 15: Common Chemical Industries in Pakistan

### Exercise Q1 — Fill in the blanks (selection)

Agriculture has been one of the oldest industries known to man since 5000 B.C. in China.

Pakistan is rich in minerals and raw materials for chemical industries.

The fertilizer industry in Pakistan is based mainly on natural gas as raw material.

### Exercise — Short questions (selection)

Why is the chemical industry important for Pakistan?

Chemical industries convert local raw materials into useful products like fertilizers, cement and soap.

Fertilizers raise crop yields and support agriculture, the backbone of the economy.

These industries provide jobs and reduce dependence on imports.

They also supply medicines, paints and plastics needed in daily life.

Name the main raw materials used by the fertilizer industry in Pakistan.

The main raw material is natural gas, which provides hydrogen for ammonia synthesis.

Ammonia is then converted into urea, the chief nitrogenous fertilizer.

Rock phosphate is used for phosphatic fertilizers like superphosphate.

Sulphur and air are also used in the manufacture of sulphuric and nitric acids.

### Exercise Q1 — Fill in the blanks (book wording)

Fertilizers enhance the natural fertility of the soil.

Micro-nutrients are required in quantity ranging from 6 grams to 200 grams per acre.

Ammonia contains 82% nitrogen.

Manure is an organic material used to fertilize land.

Cement was first introduced by an English mason Joseph Aspdin .

Phosphorus is required to stimulate root growth of plant.

In Pakistan, bleaching of pulp is carried out with chlorine or sodium hypochlorite .

Cement is generally manufactured using wet process.

The use of cement in the construction of building is based on its property of setting to a hard mass when its paste with water is allowed to stand for sometime.

Lignin is an organic polymer and causes paper to become brittle.

**Exercise Q2 — Indicate True or False**

Potassium fertilizers are especially used for tobacco and corn.

True

Ammonia is used in gaseous state while all other fertilizers are used in the solid form.

True

In wet process for the manufacture of cement, grinding of raw material is done in the presence of water.

True

The total production of cement in Pakistan is 56,30,100 metric tons/annum.

False

In neutral sulphite semi-chemical process, sodium sulphite is used buffered with sodium carbonate.

True

Lignin is an inorganic binder.

False

Paper consumption in Pakistan is around 5kg per person per year.

True

Urea contains 90% nitrogen.

False

The temperature of the digester in paper industry should be around 160-180 C.

True

Potassium fertilizers increase the capability of plants to resist diseases.

True

## Exercise Q3 — Multiple choice questions

Which three elements are needed for the healthy growth of plants?

A. N, S, P

B. N, Ca, P

✓ C. N, P, K

D. N, K, C

Which woody raw material is used for the manufacture of paper pulp?

A. Cotton

B. Bagasse

✓ C. Poplar

D. Rice straw

The nitrogen present in some fertilizers helps plants:

A. to fight against diseases

B. to produce fat

C. to undergo photosynthesis

✓ D. to produce protein

Phosphorus helps the growth of:

✓ A. root

B. leave

C. stem

D. seed

Micro-nutrients are required in quantity ranging from:

A. 4-40g

✓ B. 6-200g

C. 6-200kg

D. 4-40kg

During the manufacturing process of cement the temperature of the decomposition zone goes up to:

A. 600 C

B. 800 C

✓ C. 1000 C

D. 1200 C

The word paper is derived from the name of which reedy plant?

A. Rose

B. Sun flower

✓ C. Papyrus

D. Water Hyacinth

Which is not a calcareous material?

A. lime

✓ B. clay

C. marble

D. marine shell

How many zones through which the charge passes in a rotary kiln?

✓ A. 4

B. 3

C. 2

D. 5

Ammonium nitrate fertilizer is not used for which crop?

A. Cotton

B. Wheat

C. Sugar cane

✓ D. Paddy rice

**Exercise Q1–Q10 (continued)**

Q4. What are phosphatic fertilizers? How are they prepared? Mention the role of phosphorus in the growth of plants.

Phosphatic fertilizers supply phosphorus to plants; the main ones are superphosphate and triple phosphate.

Superphosphate is prepared by treating phosphate rock with sulphuric acid; triple phosphate by treating phosphate rock with phosphoric acid.

Phosphorus stimulates root growth, flowering and fruiting, and is part of ATP and nucleic acids, so it is essential for energy transfer in the plant.

Q5. (a) What are fertilizers? Why are they needed? (b) Discuss the classification of fertilizers and their uses. (c) How is urea manufactured in Pakistan? Describe the process in detail.

Fertilizers are natural or artificial substances containing nutrient elements that improve the growth and productiveness of plants.

They are needed because continuous cropping removes nitrogen, phosphorus and potassium from the soil, and the soil alone cannot replace them fast enough.

They are classified as nitrogenous (urea, ammonium nitrate), phosphatic

(superphosphate, triple phosphate) and potassium fertilizers (potassium chloride, potassium sulphate).

Urea is manufactured in Pakistan from liquid ammonia and carbon dioxide: they react to give ammonium carbamate, which dehydrates to urea; the urea solution is concentrated and prilled in a prilling tower.

Q6. (a) What are the prospects of fertilizer industry in Pakistan? (b) What are essential nutrient elements and why are these needed for plant growth? (c) Write down the essential qualities of a good fertilizer.

Pakistan has large reserves of natural gas, limestone and phosphate rock, and a large agricultural sector, so the fertilizer industry has strong prospects for growth.

Essential nutrient elements are those without which a plant cannot complete its life cycle: nitrogen, phosphorus, potassium, calcium, magnesium, sulphur, and trace elements such as boron, iron and zinc.

They are needed for proteins, chlorophyll, energy transfer, cell walls and enzyme action.

A good fertilizer must be water-soluble and readily available to plants, stable in storage, cheap, of suitable concentration, and non-toxic to plants and soil.

Q7. (a) Describe the composition of a good portland cement. (b) Discuss the wet process for the manufacturing of cement. (c) What is setting of cement? Discuss the reactions taking place in the first 24 hours.

Good portland cement contains about 60-67% lime (CaO), 17-25% silica ($SiO_2$), 3-8% alumina ($Al_2O_3$), 0.5-6% iron oxide ($Fe_2O_3$), with small amounts of magnesia and sulphur trioxide.

In the wet process, limestone and clay are ground with water to a slurry, which is fed into a rotary kiln and passes through drying, preheating, decomposition and burning zones to give clinker, which is ground with gypsum to cement.

Setting of cement is the hardening of its paste with water into a hard mass.

In the first 24 hours, tricalcium aluminate hydrates rapidly and tricalcium silicate hydrates to calcium silicate hydrate gel and calcium hydroxide, giving the first set and early strength.

Q8. What are the essential non-woody raw materials used in the production of pulp and paper in Pakistan?

The essential non-woody raw materials in Pakistan are bagasse (sugarcane residue), rice straw, wheat straw, cotton stalks, and grasses such as kahi and sarkanda.

These materials are available in large quantities as agricultural waste and are the main raw materials of the Pakistani paper industry.

Their fibres are separated from lignin by pulping, then washed, bleached and formed into paper sheets.

Q9. (a) What are the principal methods of chemical pulping? (b) Describe the neutral sulphite semi-chemical process.

The principal chemical pulping methods are the sulphate (kraft) process, the sulphite process, and the soda process, with the neutral sulphite semi-chemical (NSSC) process for semi-chemical pulp.

In the NSSC process, the raw material is cooked with sodium sulphite buffered with sodium carbonate at about 160-180 C in a digester.

The cook softens the lignin without fully dissolving it, giving a strong pulp with high yield, which is then washed, refined and bleached.

Q10. (a) What are the common bleaching agents used in paper industry in Pakistan? Briefly describe the bleaching process. (b) What are the prospects of paper industry in Pakistan?

The common bleaching agents in Pakistan are chlorine, sodium hypochlorite, calcium hypochlorite and hydrogen peroxide.

In bleaching, the washed pulp is treated in stages with these agents, which oxidise and remove the remaining coloured lignin, giving white pulp.

The paper industry in Pakistan has good prospects because demand for paper, packaging and textbooks is growing, but it faces raw-material shortages and needs more recycling of waste paper.

## Chapter 16: Environmental Chemistry

### Exercise Q1 — Fill in the blanks (selection)

Acid rain is due to the presence of oxides of sulphur and nitrogen which get mixed with the rain water.

The substances which can directly kill the unwanted organisms are called pesticides .

Chlorine is frequently used to disinfect water.

The ozone layer protects the earth from harmful ultraviolet rays in the sunlight.

### Exercise Q3 — Multiple choice questions (selection)

To avoid the formation of toxic compounds with chlorine, which substance is used for disinfecting water?

A. KMnO4

✓ B. O3

C. Alums

D. Chloramines

Newspaper can be recycled again and again by how many times?

A. 2

B. 3

C. 4

✓ D. 5

## Exercise — Short questions (selection)

Q6. What is acid rain and how does it affect our environment?

Acid rain is rain containing dissolved oxides of sulphur and nitrogen as acids.

These gases come from burning fossil fuels in factories, vehicles and power stations.

Acid rain damages crops, forests and buildings made of marble and limestone.

It also acidifies lakes and rivers, killing fish and other aquatic life.

Q7. What is smog? Explain the pollutants which are the main cause of photochemical smog.

Smog is a mixture of smoke and fog that hangs over cities and reduces visibility.

Photochemical smog forms when nitrogen oxides and hydrocarbons react in sunlight.

Ozone, PAN and aldehydes produced in this reaction irritate the eyes and lungs.

It also damages plants and rubber goods, and causes breathing difficulties.

Q8. Why is the ozone layer depleting? What will happen when its concentration decreases?

Chlorofluorocarbons released from refrigerators and sprays rise to the stratosphere.

Ultraviolet light breaks them down, releasing chlorine atoms that destroy ozone.

One chlorine atom can destroy thousands of ozone molecules.

Less ozone means more harmful ultraviolet rays reach the earth, causing skin cancer and harming crops.

Q14. How is water purified, i.e., made potable? Discuss in detail.

Raw water is first screened to remove large solids, then left in settling tanks.

Alum is added to coagulate fine particles, which settle as sludge.

The water is filtered through sand beds to remove remaining suspended matter.

Finally it is disinfected with chlorine or ozone to kill germs, making it safe to drink.

## Exercise Q1 — Fill in the blanks (continued)

Only 1% of the total earth's water resources are available as fresh water.

An ecosystem is a smaller unit of biosphere which consists of community of organisms and their interaction with environment.

Carbon monoxide binds blood haemoglobin 200 times more strongly than oxygen thus excluding oxygen from normal respiration.

The elevated concentration of suspended solids is harmful for fish as it clogs the gills thus causing suffocation.

The ozone layer in the stratosphere surrounds the globe and filters most of the harmful UV rays in the sunlight before they could reach the earth.

The presence of pathogenic microorganisms in livestock waste can contaminate surface and ground water causing various infectious diseases.

Incineration is not a clean process because it produces air pollution and toxic ash .

A process in which some of the used or waste materials are not discarded after their initial use but are processed so that it can be used again is called recycling .

**Exercise Q2 — Indicate True or False**

Half of the mass of the atmosphere is concentrated in lower 10 km.

True

The oceans cover approximately 71 percent of the earth.

True

The volcanoes produce 55% of SO2.

False

The reducing smog is due to the presence of nitric oxide.

False

Ozone is produced in the polar regions by the photochemical reaction of oxygen.

False

The temperature in the troposphere decreases with the increasing altitude from 15

to -56 C.

True

Incineration is a waste treatment process in which solid waste is dumped in a land fill.

False

Acid rain is due to the presence of oxides of sulphur and nitrogen which get mixed with the rain water.

True

The heavy metals have a safe limit where they are not toxic.

True

The reprocessing of the plastics is to convert back to their components by a chemical or thermal process so that these can be used again.

False

**Exercise Q3 — Multiple choice questions (continued)**

The pH range of the acid rain is:

A. 7-6.5

B. 6.5-6

C. 6-5.6

✓ D. less than 5

Peroxyacetylnitrate (PAN) is an irritant to human beings and it affects:

✓ A. eyes

B. ears

C. stomach

D. nose

A single chloride free radical can destroy how many ozone molecules?

A. 100

✓ B. 100000

C. 10000

D. 10

Fungicides are the pesticides which:

✓ A. control the growth of fungus

B. kill insects

C. kill plants

D. kill herbs

Ecosystem is a smaller unit of:

A. lithosphere

B. hydrosphere

C. atmosphere

✓ D. biosphere

The main pollutant of leather tanneries in the waste water is due to the salt of:

A. lead

✓ B. chromium (VI)

C. copper

D. chromium (III)

In purification of potable water the coagulant used is:

A. nickel sulphate

B. copper sulphate

C. barium sulphate

✓ D. alum

The temperature in the non-rotating chamber in the incineration of industrial and hazardous waste process has a range:

A. 900 to 1000 C

B. 250 to 500 C

✓ C. 950 to 1300 C

D. 500 to 900 C

**Exercise Q1–Q5, Q9–Q10, Q12–Q13, Q15–Q16 (Q6, Q7, Q8, Q14 already present)**

Q4. Discuss in detail the components of the environment.

The environment consists of four components: the atmosphere (gases around the earth), the hydrosphere (all water), the lithosphere (rocks and soil), and the biosphere (all living things).

These components interact: air, water, soil and life exchange matter and energy continuously.

Environmental chemistry studies the sources, reactions, transport and effects of pollutants in these components.

Q5. Describe the natural and human sources of carbon monoxide, nitrogen oxides and sulphur oxides.

Carbon monoxide comes naturally from volcanoes and forest fires, and from human sources such as automobile exhaust, incomplete combustion of fuels and industrial furnaces.

Nitrogen oxides come naturally from lightning and bacterial action, and from vehicles, power stations and furnaces burning fuel at high temperature.

Sulphur oxides come naturally from volcanoes and decaying organic matter, and from burning coal and oil containing sulphur, smelters and refineries.

Q9. How is oil spillage affecting the marine life?

Oil spilled on the sea forms a layer on the surface that blocks light and oxygen from reaching marine plants and animals.

Birds and mammals coated with oil lose insulation and buoyancy, and many die from poisoning when they clean themselves.

Oil also damages beaches, fishing grounds and coral reefs, and its toxic components poison fish and shellfish for years.

Q10. How are detergents a threat to aquatic animal life?

Detergents contain phosphates and surfactants that wash into rivers and lakes. Phosphates cause algal blooms, which use up dissolved oxygen when they decay, suffocating fish and other aquatic animals.

Surfactants damage the gills and cell membranes of aquatic animals, and detergents mobilise bound toxic metals from sediments into the water.

Q12. Explain how pesticides are dangerous to human beings.

Pesticides are toxic chemicals that kill pests, but they also poison farm workers, contaminate food and drinking water, and accumulate in the food chain.

They can damage the nervous system, liver and kidneys, cause cancer and birth defects, and kill useful insects, birds and fish.

Persistent pesticides such as DDT remain in soil and water for years and build up in the fatty tissues of animals and humans.

Q13. Discuss industrial waste effluents.

Industrial waste effluents are liquid wastes discharged by factories, containing acids, alkalis, heavy metals, oils, dyes and toxic organic chemicals.

When discharged untreated into rivers, they kill aquatic life, make water unfit for drinking and irrigation, and contaminate soil and groundwater.

They must be treated before discharge: neutralised, settled, filtered and biologically treated, and heavy metals must be removed.

Q15. What are leachates?

Leachates are liquids formed when water percolates through solid waste in a landfill, dissolving and suspending contaminants.

They contain dissolved organic matter, ammonia, hydrogen sulphide, heavy metals and disease-causing microorganisms.

If a landfill is not properly lined, leachate leaks into soil and groundwater and contaminates drinking water supplies.

Q16. Explain the process of incineration of industrial waste.

Industrial and hazardous waste is incinerated at high temperature in a rotary kiln, which accepts liquid, solid or sludge waste.

The waste burns in a rotating chamber, and the gases pass to a non-rotating (secondary) chamber kept at 950-1300 C, where combustion is completed.

The heat may be used to raise steam, and the remaining ash is disposed of in a landfill; incineration is not clean, because it produces air pollution, toxic ash and dioxins.

Q11 — flagged, not located in the extracted exercise pages.

BOOK NUMBERING NOTE (render-verified): book p.28 lists Q.10 and then Q.12-Q.16, with no Q11 printed. This is a book numbering skip, not an omitted question. Q6 (acid rain), Q7 (smog), Q8 (ozone) and Q14 (water purification) are already present in the existing notes and are not duplicated here.